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Where the Bytes Go: Certificate Chains, Not Key Encapsulation, Dominate Post-Quantum Handshake Cost on Unmanned Aircraft Command-and-Control Links

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Abstract

Post-quantum migration of UAV command-and-control (C2) links is usually assessed by timing ML-KEM and ML-DSA on a flight controller. We argue this measures the wrong quantity: on a C2 link the binding constraint is bandwidth. Identity dominance follows from FIPS 203/204 sizes; whether a stack reproduces it, and its cost in seconds, does not. On TLS 1.3 over TCP, on an emulated 9.6 kbps/250 ms link with mutual authentication and a self-signed, depth-one chain, 18,310 of the 20,840 bytes an ML-KEM-768 + ML-DSA-65 handshake sends are certificate, signature and the records carrying them — 7.2 times the key exchange on the wire, 7.8 on the standards objects alone, with the wire agreeing to the byte. FN-DSA-512, extrapolated along the measured slope, gains only 12–13 % over ML-DSA-44: not worth a draft standard. One intermediate certificate costs 11,248 bytes, more than any signature change saves. The penalty is neither rate- nor delay-invariant: 8.64× to 3.07× across 2.4–38.4 kbps, 6.7× at 40 ms low-Earth orbit, so the geostationary figures are conservative. Removing certificates beats shrinking them: a pre-shared-key handshake carries 2,902 bytes and takes 3.32 s, 1.67× the classical time, and loss does not reorder them. Five measurement pitfalls documented, one inflating the count by 54.7 %. Readiness is an identity-architecture decision, not an algorithm choice.

Keywords: post-quantum cryptography; ML-KEM; ML-DSA; TLS 1.3; unmanned aerial vehicles; command and control; MAVLink; certificate chains; pre-shared keys; performance measurement

1. Introduction

Unmanned aerial vehicle command-and-control links carry telemetry, waypoint uplinks and mission commands over radio paths that are, by design, reachable from the ground. Most deployed links protect that traffic with elliptic-curve key establishment and, where message authentication exists at all, with symmetric or elliptic-curve signatures. Both are broken by Shor's algorithm [1]. Because the operational lifetime of an airframe exceeds the plausible horizon for a cryptographically relevant quantum computer, recorded C2 traffic is a harvest-now-decrypt-later target, and forged command injection becomes a safety issue rather than merely a confidentiality one.

The NIST post-quantum standards remove the algorithmic obstacle: FIPS 203 [2] specifies ML-KEM, FIPS 204 [3] specifies ML-DSA, and FIPS 205 [4] specifies SLH-DSA. They are final, they have X.509 encodings [5,6], and they are implemented in mainstream libraries including OpenSSL 3.5 [7]. What remains is an engineering question: what does it cost to put these primitives on the link.

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The prevailing way to answer that question is to benchmark the primitives on the target microprocessor. A substantial literature reports cycle counts, stack usage and energy for ML-KEM and ML-DSA on Cortex-M4-class parts [8,9], and the numbers are useful for deciding whether a given flight controller can afford the computation. But a flight controller can almost always afford the computation. What it cannot afford is the airtime. On a 9.6 kbps beyond-line-of-sight satellite link, the cryptographic computation in a ML-KEM-768 + ML-DSA-65 handshake occupies tens of milliseconds on a Cortex-M4-class flight controller — the cycle counts in [8,9] put two key generations, two encapsulations, two signatures and two verifications in that range — while the bytes occupy eleven seconds. Optimising the first term cannot move a quantity that the second term dominates by two to three orders of magnitude. That premise is usually left as a citation; we also measure the arithmetic, on the machine that runs the testbed, and report the margin it leaves (Section 3.4). The margin is three orders of magnitude, so the premise holds for any processor within that distance of a desktop core rather than only for the one quoted.

This paper therefore asks a different question: *which* bytes. A post-quantum TLS handshake contains three kinds of payload — the key exchange, the certificate, and the signature over the handshake transcript — and the standards documents make all three sizes public. The split is therefore predictable, and it is worth saying so plainly rather than presenting it as a discovery: two ML-DSA-65 certificates of 5,514 bytes and two signatures of 3,309 against 2,272 bytes of ML-KEM public key and ciphertext is 7.8 to 1 before a packet is sent. What the standards do not give is whether a production stack reproduces that arithmetic, what the split costs in seconds on a link where seconds are the budget, and which of the available remedies is worth most. We measure the first, convert it into the second, and rank the third. The ratio on the wire is 7.2 to 1 in favour of identity material — lower than the object ratio, because framing inflates the smaller term faster, as Section 4.2 accounts for — and it reorganises the problem: the migration cost is set by the public-key infrastructure, not by the lattice arithmetic.

The distinction matters because the two halves have different remedies. Key exchange has no cheaper form worth taking: ML-KEM-768 is already the compact standardised option, and moving to ML-KEM-1024 costs little more. Identity material, by contrast, has three quite different remedies — use a smaller signature scheme, use a shorter chain, or do away with certificates altogether — and at 2.25, 4.94 and 7.97 s respectively they differ in value by a factor of three and a half. We quantify all three, and the last dominates.

Our contributions are:

- A byte-level attribution of the post-quantum mutual-authentication handshake, measured on a real TLS 1.3 stack rather than derived from object-size tables, showing that identity material accounts for 87.9 % of the exchange (Section 4.1).
- Three controlled sweeps: one over ML-DSA-44/65/87 that yields the marginal throughput of the link (2,150–2,282 B/s at 9.6 kbps, 90–95 % of the ideal serial rate) and lets a byte count be converted into a time on links that cannot be measured directly; one over link rate from 2.4 to 38.4 kbps at fixed propagation delay, which shows that the certificate penalty runs from 8.64× to 3.07× and is therefore not a constant; and one over one-way delay from 0 to 600 ms, which shows that the ratio falls to 4.72× on a high-latency path while the absolute penalty grows from 8.80 to 9.99 s, so the geostationary figure this paper leads with is the conservative end of the range (Sections 4.3, 4.4 and 4.5).
- A measurement of pre-provisioned trust, showing that replacing the certificate exchange with a TLS 1.3 external pre-shared key reduces the handshake from 20,840 to 2,902 bytes and from 11.29 to 3.32 s, bringing post-quantum C2 within 1.67× of its classical cost (Section 4.6).

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- Two measurements of remedies that the literature usually assumes rather than measures. A second certificate in the chain costs 11,248 bytes and 4.94 s, a factor of 1.44 — more than the best signature-algorithm change buys — and the figure resolves to the byte against the standard (Section 5.3). Session resumption costs 0.27 of the bytes and 0.31 of the time of a cold handshake but is *not* cheaper than a raw pre-shared key (3,335 against 2,902 bytes), and in OpenSSL 3.5.7 is unavailable at all when client authentication is requested, because its server issues no ticket in that mode (Section 5.2).
- An assessment of what compact signatures can and cannot buy: extrapolating along the measured slope, FN-DSA-512 gains only 12.6 % over ML-DSA-44, which does not pay for a draft standard, a 2.3× slower signature and 40–57 kB of expanded key state (Section 4.8).
- A documented account of five measurement pitfalls that each corrupted a headline number in our own first pass, including one that inflated the handshake by 54.7 % and one that made the emulated link run 53 % fast (Section 3.3). We report these because the failure modes are silent: every one of them produced plausible numbers.

Section 2 fixes notation and reviews prior work. Section 3 describes the testbed, validates the link emulation, documents the pitfalls, and states the scope (Section 3.4). Section 4 reports the measurements. Section 5 turns them into design rules and states what the study does not cover.

2. Background and Related Work

2.1. Object Sizes in the NIST Standards

Table 1 collects the public-key and signature sizes that the rest of the paper relies on. The security categories were fixed in advance of the selection [10], and the schemes themselves are surveyed in [11,12]. Two facts drive everything downstream. First, ML-KEM ciphertexts and public keys are around a kilobyte, which is large next to X25519's 32 bytes but small next to a certificate. Second, ML-DSA signatures are 2.4–4.6 kB and public keys 1.3–2.6 kB, so a certificate that carries both is 4–7 kB before any name, validity or extension fields are added. The asymmetry between the two columns is the subject of this paper.

Table 1. Object sizes from the final NIST standards, in bytes. These are exact values from FIPS 203/204/205; the measurement in Section 4 confirms that the on-the-wire sizes follow from them plus the X.509 and TLS framing overheads.

Scheme	Parameter set	Public key	Ciphertext	Signature
ML-KEM	768	1,184	1,088	–
ML-KEM	1024	1,568	1,568	–
ML-DSA	44	1,312	–	2,420
ML-DSA	65	1,952	–	3,309
ML-DSA	87	2,592	–	4,627
SLH-DSA	128s	32	–	7,856
FN-DSA	512	897	–	666
X25519 / ECDSA-P256	–	32 / 64	32	64

2.2. The UAV C2 Link

Two profiles bracket the problem. Line-of-sight operations use a telemetry radio or a Wi-Fi or LTE data link, with rates from tens of kbps to tens of Mbps and round-trip times of a few milliseconds. Beyond-line-of-sight operations in controlled airspace use a satellite C2

link of the kind anticipated by DO-362 [13]: roughly 9.6 kbps and a one-way propagation delay of 250 ms, giving a round-trip time of half a second. The second profile is the one that constrains design, and it is the one we emulate.

UAV networking at this layer has been surveyed extensively [14–16]; what matters here is narrower, namely the byte budget of one handshake. Two protocol families carry the traffic. MAVLink [17] is the de facto autopilot protocol; its signed-frame mode authenticates individual frames with a symmetric MAC, and its payload limit is 255 bytes, so any object larger than a few hundred bytes must be fragmented by the application. Alternatively the link can carry a datagram transport and run DTLS [18,19] over it, in which case the transport performs the segmentation. The choice between these two encapsulations is a design decision with measurable consequences, and it is orthogonal to the choice of cryptographic algorithm.

Either way a post-quantum certificate does not fit in one MAVLink frame: a MAVLink v2 frame carries at most 255 bytes of payload, with 12 bytes of header and checksum outside it, so a 5,549-byte certificate spans twenty-two frames and the receiver has to hold all 5.4 kB of them before it can verify anything — 2.8 % of the 192 kB floor on a typical flight controller, which is affordable, but it is buffered before a single byte of the handshake has been authenticated. That is not an argument against MAVLink signing, which protects frames rather than sessions, but it is why the measurements here are taken at the transport layer: any session-level post-quantum authentication on a MAVLink link will be carried by DTLS or by a TLS tunnel, and the byte budget is set there.

2.3. Related Work

Work on post-quantum cryptography for UAVs has concentrated on three things: making the primitives fit an embedded budget, integrating them into MAVLink, and authenticating the link.

On the first, the pqm4 library [8] and its successors report cycle counts and memory for the NIST candidates on Cortex-M4, and [9] extends the analysis to energy. These results establish that the primitives are affordable on a flight controller. They do not address airtime, and on the links that matter the computation is not the binding constraint.

On the second, [20] proposes a lightweight mutual-authentication and encryption scheme combining Kyber key exchange with ASCON-AEAD for MAVLink message protection, reporting roughly 2.1 ms of processing time in a simulated UAV environment. That work optimises computation and reports computation; the byte cost on a narrow link is not its subject.

The closest UAV work in this respect is [21], a Kyber-based identity authentication and key agreement scheme for UAV networks. Its authors capture the handshake with Wireshark and report 1,836 bytes of communication overhead, 42 % below the Kyber.AKE baseline and 54 % below the DTLS baseline they compare against. Three properties separate that number from ours. It is a single capture with no repetition, so it carries no variance. It is a total rather than a breakdown, so it cannot say whether the bytes sit in key exchange or in identity material. And 1,836 bytes is 4.1 % *below* our classical X25519+ECDSA-P256 baseline of 1,915 bytes (Table 5) — which is what a handshake costs when no certificate chain is on the wire. A pre-provisioned design is one explanation for that figure, though a single total cannot distinguish it from a handshake that simply carried a shorter chain, and we claim only the first reading. Read that way, the one published UAV byte count we can compare against is consistent with the thesis we test below: removing the chain dominates every choice made inside it.

Industrial designs reach a related conclusion from the other direction, though they are vendor documentation rather than peer-reviewed studies: the MORVS Swarm SDK

0.4.0 release notes state that post-quantum signatures are not implemented at all, that ML-KEM-768 is opt-in and used for session setup only, and that its 1,088-byte ciphertext travels with the first message [22]. A sufficient explanation is arithmetic that we perform here rather than quote: an ML-DSA-87 signature is 4,627 bytes, and a MAVLink v2 frame carries at most 255 bytes of payload, so one signature alone spans nineteen frames.

MAVLink v2 signing deserves a word here, because it is sometimes offered as an answer to this problem. It appends a 48-bit keyed hash computed with a secret shared out of band, which detects corrupted and injected frames but establishes no peer identity and derives no session key: it is a frame-integrity check, not a substitute for the handshake. The two layers are also ordered the other way round from what the objection assumes — TLS runs beneath MAVLink, so a MAVLink frame is application payload inside a TLS record, not a container for one. What is worth noting is that MAVLink signing *already* assumes a pre-provisioned symmetric secret at both ends, which is precisely the trust model Section 4.6 measures. The UAV ecosystem has made that decision once, at the frame layer; the argument below is that it extends to the handshake layer for the same reasons.

Outside the UAV literature, the size problem in post-quantum TLS is well recognised and has been quantified at the PKI layer. The IETF Merkle Tree Certificate draft — an Internet-Draft, not yet a standard — counts the authentication overhead of a single ML-DSA-65 leaf — certificate, two signed certificate timestamps and the leaf signature — at 9,927 bytes, and argues that this changes the economics of public logging rather than merely the size of one message [23]. For a two-level chain the figure is larger still: in our own testbed one ML-DSA-65 certificate occupies 5,549 bytes of TLS `certificate_list` (Table 3), so a leaf plus an intermediate reaches 11,093 bytes in each direction before any handshake signature is counted. That is the regime in which the initial congestion window of RFC 6928 [24] (about 14.6 kB) stops absorbing the handshake. The industrial response has been structural rather than algorithmic: Let's Encrypt announced in June 2026 that it would pursue Merkle Tree certificates rather than issue conventional ML-DSA certificates [25], on the grounds that naive substitution does not fit the wire budget at web scale.

Measured post-quantum TLS is a literature of its own, and it is where the byte question was first raised. Sikeridis et al. evaluate the NIST signature candidates in TLS 1.3, reporting both handshake latency and server throughput [26], and extend the accounting to SSH [27]; Paquin, Stebila and Tamvada benchmark key exchange and authentication in TLS under realistic network conditions [28]. Both lines conclude that post-quantum authentication is affordable on the public Internet, and both locate the cost in the signature and the certificate rather than in the key exchange. Our result is the same effect under a different constraint, and it should be read that way: where those studies ask whether a server can afford the extra milliseconds, we ask what happens when the link, not the CPU, is the budget. Schwabe, Stebila and Wiggers make the structural version of the argument, replacing the server's handshake signature with KEM-based authentication and reporting less than half the bandwidth of a size-optimised post-quantum TLS [29]. That is the closest prior art to our recommendation, and the difference is worth stating precisely: KEMTLS removes the server's signature in favour of a KEM public key, but the client still signs, and both parties still need long-term key material with a distribution and revocation story. The pre-shared key removes the signature and the certificate together, and in our measurement the certificate is where 87.9 % of the bytes sit.

Our work differs from all three strands. Relative to the UAV literature, we measure the link rather than the microprocessor, and we separate the contributions of key exchange and identity material instead of treating the handshake as one object. Relative to the TLS-level measurements, we work at a bandwidth three orders of magnitude below theirs and on a link whose budget is airtime rather than latency, which changes which remedy wins: a few

hundred milliseconds is a rounding error on the public Internet and eleven seconds is a mission constraint here. Relative to the web-PKI analyses, we observe that a UAV C2 link is *not* the web: an operator who owns both endpoints can pre-provision key material and skip certificates entirely, an option a public certificate authority cannot offer. The measurement below puts a number on that option, and the number is large.

3. Measurement Methodology

3.1. Testbed

All measurements use OpenSSL 3.5.7 [7], which implements ML-KEM and ML-DSA natively and negotiates them as TLS 1.3 [30] named groups and signature algorithms. No external provider is involved, so the byte counts and timings are those of a production stack rather than of a research prototype. The negotiated cipher suite is `TLS_AES_256_GCM_SHA384` throughout, which is what fixes the 16-byte AEAD tag and the 48-byte `Finished` that the byte accounting below relies on. Key exchange uses the pure ML-KEM-768 group; the hybrid group that browsers now negotiate by default would add the 32-byte X25519 public key and nothing else, which does not change any conclusion here. Research libraries such as liboqs [31] offer a wider algorithm set — including the Falcon implementation we would have needed for Section 4.8 — but they do not implement the TLS record layer, and the record layer is what this paper measures.

A client and a server process run on the same host and communicate through a userspace TCP proxy that rate-limits and delays the connection. The proxy applies a token bucket in each direction independently, followed by a fixed one-way propagation delay per forwarded chunk. Independent per-direction rate limiting matches the uplink and downlink of a real C2 radio, which are separate channels. Because chunks are forwarded strictly in order, the per-chunk delay composes into correct store-and-forward behaviour without requiring `tc`, root privileges or dedicated hardware. That composition is an argument rather than a validation: what was checked directly against a known transfer is the serialisation rate, to $+0.3\,\%$, and Section 4.5 reads only ratios between configurations from the delay axis for that reason.

Authentication is mutual throughout, because a C2 link authenticates the aircraft to the ground station and the ground station to the aircraft. The server requests a client certificate and the client presents the same self-signed certificate it uses as its own identity, so both directions carry a certificate and a `CertificateVerify` signature of the same algorithm. This is the correct model for the deployment and, as Section 3.3 explains, it is also the configuration in which one of the measurement artefacts appears.

The client sends a single byte once the handshake has completed, as a liveness probe; the server returns no application data. That byte occupies a 23-byte record — 5 of record header, 16 of AEAD tag, one of content type and one of payload — which the byte counter includes. Every total reported here is therefore 23 bytes larger than the handshake proper, and Table 7 names the record so the convention can be undone by hand. Session tickets are disabled on both sides.

3.2. Link Emulation and Its Validation

The proxy was validated against a known transfer before any handshake was measured. Pushing 12,000 bytes through the throttle at a 9.6 kbps target yields 9.57 kbps, an error of $+0.3\,\%$. We report this because two earlier versions of the throttle were wrong in ways that were not visible in the handshake numbers themselves; both are described in Section 3.3.

The emulation has two properties that matter for interpretation and that we state explicitly rather than bury.

First, the configured one-way delay is applied once per forwarded chunk, not once per link hop. That is not the same thing as store-and-forward, and the difference is measurable rather than rhetorical: on the 20,840-byte handshake at 9.6 kbps, completion takes 10.26 s with zero delay and 11.28 s with 250 ms, so repeated delay contributes 1.0 s, or 9 % of the total. The other 91 % is serialisation. The direction of the resulting error is conservative: a real link pays the delay once per direction change, and on a narrow link TCP coalesces a flight of segments into one chunk, so chunk boundaries tend to coincide with the direction changes where a delay is genuinely owed. We therefore report the propagation term as an upper bound of at most a handful of delay periods, which at 250 ms is under a second on the largest handshake. That argument is an argument, not a measurement: we did not sweep the chunk size, because the proxy forwards whatever `recv` returns rather than a configured block, so the number of delay payments follows the sender's segmentation rather than a parameter of ours. A path that charged delay per smaller unit would pay more of them, which moves the absolute times up and leaves the ratio between configurations where it is; Section 5.6 records this as a limitation.

Second, the round-trip time is the sum of the configured propagation delay and the serialisation time of whatever is in flight. At 9.6 kbps a 1,200-byte chunk takes a full second to serialise, so "round-trip time" is not a constant in this regime — it is dominated by the transfer size. This is a property of real narrowband links as well, not an artefact of the emulator.

3.3. Five Pitfalls, and What Each One Cost

Every number reported in Section 4 was wrong at least once before it was right. We document the failure modes because each produced a plausible value, and because three of them are specific to measuring on a real stack rather than to modelling. Table 2 is the summary; the paragraphs that follow give the derivation of each, since the numbers are only credible with it.

Table 2. The five measurement pitfalls, and what each one cost before it was found. The last column is the change that removed the error; no number reported in Sections 4 and 5 predates these fixes.

#	Failure mode	Effect on the reported number	Correction
1	Token bucket credited the same wait twice	link ran at 14.73 instead of 9.60 kbps, 53 % fast	balance allowed to go negative
2	Free initial burst of 1,400 bytes	1.17 s uncharged at 9.6 kbps	burst set to zero
3	`-www` status page counted as handshake	more than half the classical figure	server run without `-www`
4	`NewSessionTicket` inside or outside the window	12 % byte variation between runs	tickets disabled on both sides
5	`-CAfile` certificate copied into each ticket	total 20,840 → 32,236 bytes, +54.7 %	`-no_ticket` on the server as well

(1) The token bucket credited the same wait twice. The first throttle clamped the token balance to zero when a chunk exceeded the available credit, then refilled from elapsed time on the next call. The wait already served was thus counted again as available capacity, and the link ran at 14.73 kbps against a 9.6 kbps target — 53 % fast. The fix is to let the balance go negative, so that the bucket owes capacity rather than forgetting the debt. Every timing in our first pass was compressed by this error.

(2) The token bucket handed out a free burst. With a 1,400-byte initial burst, a short transfer received its first 1,400 bytes at no cost. At 9.6 kbps that is 1.17 s, which is not negligible when the whole handshake is a few seconds. The burst is now zero, making the serialisation exact.

(3) The server's status page was counted as handshake. An early version drove `s_server` with `-www`, whose HTML status page is about 5 kB and was counted in the byte total. On the classical configuration this was more than half the reported figure: about 5 kB against a 1.9 kB handshake, 72 % of what was being counted. The server now runs without `-www` and the application payload is one byte.

(4) Session tickets were counted inconsistently. `NewSessionTicket` arrives after the handshake completes, and whether it fell inside the measurement window depended on link speed. The same configuration varied by 12 % in byte count between runs. Tickets are now disabled.

(5) The certificate authority certificate was embedded in every ticket. This is the one that matters most, because it inflated the result we most wanted to be true. With client authentication enabled, `s_server` uses its `-CAfile` both to validate the client certificate and to build the chain it sends. The CA certificate therefore reappeared inside each `NewSessionTicket`. Parsing the record layer makes it visible (Figure 1 and Table 3): with tickets disabled the two post-handshake records are 79 bytes each, which is what a ticket should cost; with the server-side flag omitted the same two records become 5,775 bytes each, 5,696 bytes larger — the size of the certificate plus ticket framing. The server to client direction grew from 10,468 to 21,860 bytes, and the handshake total from 20,840 to 32,236 bytes, an inflation of 54.7 %. The client's flight gains four bytes at the same time, and those are not an artefact: with tickets enabled the client offers the empty `session_ticket` extension, which `-no_ticket` suppresses, so its `ClientHello` grows from 1,357 to 1,361 bytes. The two directions therefore grow by 11,392 and 4 bytes, and the 11,396-byte growth of the total is their sum. Client-side `-no_ticket` does not suppress it, because the ticket is emitted by the server after the handshake. The server must be given the flag as well.

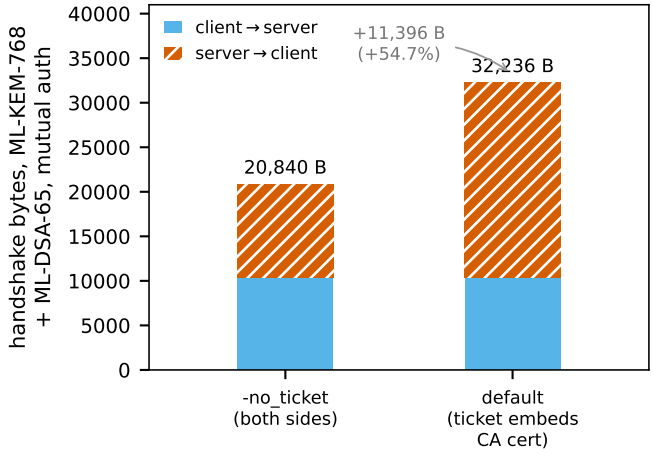


Figure 1. The ticket artefact of Section 3.3, item (5). Both bars are the same ML-KEM-768 + ML-DSA-65 mutual-authentication handshake. In the left bar the server was started with `-no_ticket` and the two post-handshake records are 79 bytes each. In the right bar that flag was omitted, and because `-CAfile` is also used to build the outgoing chain, the CA certificate is embedded in each ticket, adding 5,696 bytes twice. Reporting the right-hand bar as the cost of post-quantum authentication would have overstated it by 54.7 %.

Table 3. Record-level view of the artefact in Section 3.3, item (5), for ML-KEM-768 + ML-DSA-65. Server to client records are listed in transmission order, so the last three entries of each row are the post-handshake records. Only these three differ between the two rows. The client to server flight is not shown and grows by four bytes — 10,372 to 10,376 — which is the empty `session_ticket` extension the client offers only when tickets are enabled. The two flights therefore account for 11,392 and 4 of the 11,396-byte total.

Mode	Server→client record sizes (bytes)	Total
mutual auth, `-no_ticket`	1,183 6 28 107 5,549 3,339 74 79 79 24	10,468
mutual auth, tickets enabled	1,183 6 28 107 5,549 3,339 74 5,775 5,775 24	21,860

Pitfall (5) is worth dwelling on for a reason beyond bookkeeping. It pushed in the direction of our hypothesis, and a less careful investigation would have kept it. The published literature on PQC certificate chains reports figures in this range, and a 32 kB handshake is not obviously wrong. What caught it was not intuition but arithmetic: the 5,775-byte records are not explicable from any field in the handshake, and once a record cannot be attributed it has to be investigated rather than averaged in.

3.4. Scope: What Is Measured and What Is Not

Transport. Every measurement here runs TLS 1.3 over TCP, whereas UAV C2 links more often carry DTLS over UDP. The choice is worth stating precisely, because it is the first question a reviewer should ask.

The reason for TCP is not convenience: on this stack a post-quantum DTLS handshake cannot be run at all, and the reasons are worth stating because they are properties of the protocols rather than of our testbed. DTLS 1.3 is specified in RFC 9147 but is not implemented in OpenSSL 3.5.7, which offers DTLS 1.0 and 1.2 only. DTLS 1.2 cannot carry the handshake either, for two independent reasons. First, a TLS 1.2 cipher-suite name encodes its authentication algorithm (`ECDHE-ECDSA-...`, `ECDHE-RSA-...`) and no suite exists for ML-DSA, so an ML-DSA certificate cannot authenticate a 1.2-era handshake; measured, `s_server -dtls1_2` holding an ML-DSA-65 certificate answers every `ClientHello` with *no shared cipher*. Second, the 1.2 key exchange is ECDHE over an elliptic curve and ML-KEM is not a 1.2 group; measured, `-groups MLKEM768 -dtls1_2` fails the same way even with an ECDSA certificate. Post-quantum DTLS therefore means DTLS 1.3, and on a stack that does not implement it the choice is between measuring TLS 1.3 over TCP and measuring nothing.

What that costs can be bounded rather than guessed, because RFC 9147 changes framing only. Its record header is 13 bytes against 5, its handshake header 12 against 4, and the AEAD tag is unchanged, so a datagram of M bytes carries $M - 42$ bytes of handshake body. Applying that to the measured record list — five fragments for each 5,549-byte `Certificate` at a 1,200-byte datagram, three for each `CertificateVerify`, two for each hello — gives 21,659 bytes against 20,817, an increase of 842 bytes or 4.0 % (`harness/dtls_bound.py`; at a 1,000-byte datagram it is 1,010 bytes, 4.9 %). The certificate and signature records grow by 3.2 %, so their share of the handshake moves from 85.4 to 84.7 %: the transport scales the handshake without reordering it.

Two further effects point the same way. TCP repairs a lost segment by retransmitting that segment, whereas DTLS repairs a lost datagram by retransmitting the whole flight it belonged to, and post-quantum flights are necessarily large. And the numbers above are framing alone; a real datagram path adds IP and UDP headers that the TLS counts here do not include.

What survives intact is the attribution. RFC 9147 reuses the TLS 1.3 handshake messages unchanged, so the split between key exchange and identity material — the

paper's main result — is a property of the handshake, not of the transport that carries it. The end-to-end times are the part that would move, and they would move upward by roughly the 4 % above.

This bound runs along the transport dimension only, and holds loss equal: at a given loss rate, DTLS would cost more than the numbers here, not less. It does not cancel the opposite caveat in Section 5.6, where the emulated link is argued to be cleaner than a real one. The two act in different directions and neither is small; what the paper claims is the attribution and the ordering of remedies, which survive both.

The asymmetry also fixes the sign of that error, which is the part that matters for the ordering. The flight that carries the identity material is about 10,400 bytes, nine datagrams at the 1,200-byte MTU used above, whereas the pre-shared-key handshake completes in one or two datagrams per flight. A DTLS peer that loses a datagram retransmits its whole flight, so at equal loss probability the certificate handshake pays several times the retransmission the pre-shared-key one does. Section 4.7 measures the certificate penalty growing by 0.446 s per injected loss on TCP; on DTLS it would grow faster while the pre-shared-key penalty barely moved, and the gap between the two would widen rather than close. The ordering reported here is therefore the conservative one on the transport UAV links actually use, and the same is true of the bursty fades that Section 5.6 says are not modelled: a fade that swallows several datagrams at once costs a large flight several whole retransmissions and a small one a single datagram.

Measured: bytes in each direction, handshake completion time, and per-record attribution, all on a real TLS 1.3 stack. Also measured: the effect of changing the certificate signature algorithm while holding the key exchange fixed, the effect of replacing certificates with a pre-shared key, and how both move when the link rate varies from 2.4 to 38.4 kbps and the one-way delay from 0 to 600 ms.

Not measured: microprocessor cycle counts, RAM footprints and energy on the flight controller itself. What is measured is the arithmetic on the host that runs the testbed, and the margin it implies. A mutual-authentication ML-DSA-65 handshake performs two signatures, two verifications and one ML-KEM-768 exchange; timed with `openssl speed` on the test host these total about 1.8 ms against 11.27 s of transmission — 0.02 %, a margin of some 6,000 (`harness/compute_cost.py`). Repeat runs on the same host moved the total by about 10 %, between 1.8 and 1.9 ms, which is why the figure is quoted to two significant digits rather than four. The heaviest scheme measured, ML-DSA-87, is about 2 ms. For computation to match transmission on this link the processor would have to be 6,000 times slower than a desktop core; at a thousand times slower the arithmetic would still be under a fifth of the handshake, and the constraint of the link would still be the link.

That is the sense in which the premise is checkable: it is not that computation is fast everywhere, but that it would have to be catastrophically slow to matter, and a device that slow could not complete a TLS handshake at all. Where such figures do bear on a design decision — the FN-DSA signing cost in Section 4.8 — they are attributed to the literature and used to argue against a design, not to support one.

Behaviour under bursty packet loss is also not measured, although independent per-segment loss is: Section 4.7 injects retransmission timeouts at rates up to one segment in five and finds that the penalty is proportional to the number of segments rather than to the handshake design. Bursty fading, in which one fade swallows several segments at once, is not modelled and is flagged in Section 5.6.

4. Results

4.1. Where the Bytes Go

Table 4 and Figure 2 attribute every byte of a mutual-authentication TLS 1.3 handshake. The attribution uses a record-layer parser rather than a total counter, so the key exchange is separated from the identity material exactly: the `ClientHello` and `ServerHello` are plaintext in TLS 1.3 and their lengths are read from the wire, and everything after them is accounted by record.

Table 4. Byte attribution of the mutual-authentication TLS 1.3 handshake, measured with a record-layer parser. "Key exchange" is the `ClientHello` and `ServerHello` in both directions. "Identity" is everything else: two certificates, two `CertificateVerify` signatures, `Finished`, and record and message framing. All configurations use ECDSA-P256 for the classical baseline and the stated scheme elsewhere. The identity column is a residual — the total minus the two hello payloads — so it also absorbs the 10 bytes of hello record headers and the 23-byte probe record of Section 3; certificates and signatures proper account for 17,646 of the 18,310. The two pre-shared-key rows carry no certificate at all, so their residual is `Finished` and framing only: the 316 bytes there is what a handshake costs after the identity material is gone, not an identity overhead, and the 45.0 % and 10.9 % shares are tabulated so the column is complete rather than because they measure identity.

Configuration	Key exchange	Identity	Total	Identity share
X25519 + ECDSA-P256	330	1,583	1,913	82.8 %
X25519 + ML-DSA-65	330	18,310	18,640	98.2 %
ML-KEM-768 + ECDSA-P256	2,530	1,585	4,115	38.5 %
ML-KEM-768 + ML-DSA-65	2,530	18,310	20,840	87.9 %
ML-KEM-768 PSK	2,586	316	2,902	10.9 %
X25519 PSK	386	316	702	45.0 %

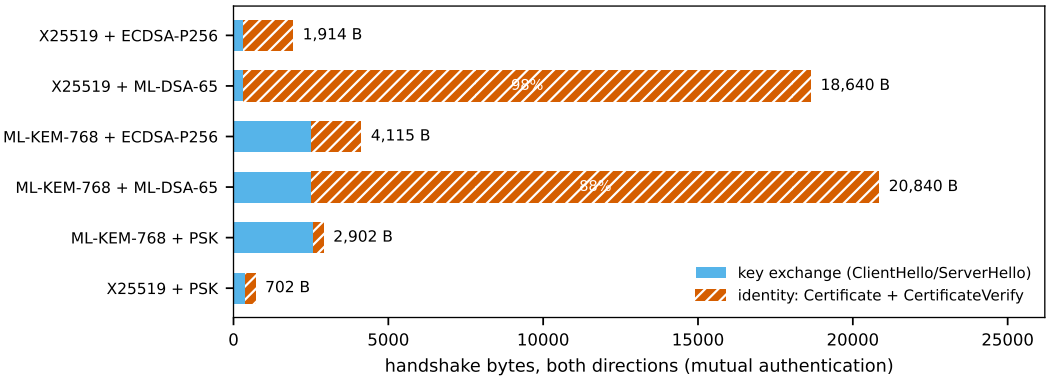


Figure 2. Byte attribution of the configurations in Table 4. The key exchange is the blue segment; the orange segment is certificate and signature material. Swapping X25519 for ML-KEM-768 adds 2,202 bytes. Swapping an ECDSA-P256 certificate for an ML-DSA-65 one adds 16,727 bytes. The post-quantum key exchange is not where the bytes are, and reducing its size is not where the leverage is.

Two of the six rows are pre-shared-key configurations, and their identity column is not identity material at all: with no certificate on the wire it holds only `Finished` and framing, which is why the share column reads 10.9 % for the post-quantum row and 45.0 % for the classical one. Those rows are there to show what remains after the identity term is removed, not to be compared with the 87.9 % above them.

The result is stark. Holding the certificate fixed and changing the key exchange from X25519 to ML-KEM-768 adds 2,202 bytes. Holding the key exchange fixed and changing the certificate from ECDSA-P256 to ML-DSA-65 adds 16,727 bytes. The ratio is 7.6 to 1.

This ratio is self-consistent in a way that gives us confidence it is a property of the protocol and not of our testbed. The identity increment is 16,727, 16,726, 16,727 and 16,727 bytes for the four key exchanges we tested (X25519, X25519ML-KEM768, ML-KEM-768, ML-KEM-1024) — constant to within one byte, as it must be if the identity material is independent of the key exchange. It also decomposes as theory requires: an ML-DSA-65 certificate is 5,514 bytes of DER against 389 for the ECDSA one, and a `CertificateVerify` signature is 3,309 bytes against 71. Doubling both for mutual authentication gives $2 \times (5{,}514 + 3{,}309) - 2 \times (389 + 71) = 16{,}726$ bytes, which is the measured column increment of Table 5 to the byte. Section 4.2 carries the same reconstruction down to individual messages.

Table 5. Full measured matrix: handshake bytes and completion time (s) for every combination of four key exchanges and four certificate signature algorithms, mutual authentication, 9.6 kbps per direction and 250 ms one-way. Cells are single runs; the repeatability of Section 4.9 puts a 1σ interval of about ± 0.02 s on each, which is smaller than the last digit shown. The row and column increments are additive to within ± 2 bytes, which is the property Section 4.1 relies on.

Key exchange	ECDSA-P256	ML-DSA-44	ML-DSA-65	ML-DSA-87
X25519	1,915 / 1.99	13,804 / 7.19	18,640 / 9.45	25,192 / 12.44
X25519ML-KEM768	4,177 / 3.86	16,068 / 9.07	20,904 / 11.33	27,456 / 14.31
ML-KEM-768	4,114 / 3.81	16,004 / 9.02	20,840 / 11.27	27,392 / 14.27
ML-KEM-1024	4,979 / 4.54	16,868 / 9.97	21,704 / 12.00	28,256 / 14.99
Column increment over ECDSA-P256, by row				
X25519	–	11,889	16,725	23,277
ML-KEM-768	–	11,890	16,726	23,278
Row increment over X25519, by column				
ML-DSA-65	2,199	2,200	2,200	2,200
ML-KEM-1024	3,064	3,064	3,064	3,064

Table 5 is the evidence for treating the two halves of the handshake as independent. Reading down a column gives the cost of the key exchange and is constant to within one byte regardless of which certificate is used; reading across a row gives the cost of the certificate and is constant to within two bytes regardless of which key exchange is used. The two byte costs add. This is what makes it legitimate to say that the certificate costs 16,726 bytes rather than that the configuration costs 20,840: the decomposition is not a modelling assumption, it is a measured invariance.

The residual ± 2 bytes is DER encoding noise, and it is classical only. ML-DSA signatures are fixed-length byte strings, so a post-quantum total is exactly reproducible: 20,840 in every run we made. The classical baseline is not, because an ECDSA signature is variable-length DER — the integers r and s gain a leading zero byte whenever the value behind them is small — so the certificate and each `CertificateVerify` move by a byte or two between runs. Table 6 collects the four values that appear in this paper and says which tables report each. Each table reports the values from the batch that produced it, and *ratios are compared only within one batch*: the certificate penalty is 5.66× in the sweep of Table 5 and 5.67× in the repeats of Table 14, a spread of 0.2 %, and no conclusion depends on which baseline is used.

Table 6. The four classical baselines in this paper, one per batch, with the tables that report each. The whole spread is four bytes, which is the DER variation described in the text; the post-quantum handshakes do not move. Ratios are formed within a batch, never across two, and the rate and delay sweeps of Tables 9 and 10 are further batches whose ratios are formed row by row, so each of their rows carries its own baseline and neither is listed here.

Batch	Reported in	Classical (B)	PQC (B)
Loss sweep, 8 runs per cell	Table 12	1,912	20,840
Attribution	Tables 4, 7; Figure 2	1,913	20,840
Matrix	Tables 5, 11, 13	1,915	20,840
Repeatability, 5 runs	Table 14	1,916	20,840

4.2. Where the 20,840 Bytes Come From

The attribution above is only credible if the individual messages can be reconstructed from the standards, so Table 7 does that. Two observations follow. First, the two directions are almost symmetric — 10,372 and 10,468 bytes — which is what mutual authentication implies, and which is why the cost of the identity material is paid twice. Second, the certificate dominates every other message in the handshake: at 5,549 bytes it is 4.7 times the next largest, and there are two of them.

Table 7. Per-message reconstruction of the measured handshake (ML-KEM-768 + ML-DSA-65, mutual authentication). Rows in italics are not messages: they collect the records that carry no handshake content, so that each direction sums to its measured total, 10,372 and 10,468. Every figure in the 'bytes' column is read off the wire by the record-layer parser; every figure in the 'contents' column is from FIPS 203/204 plus X.509 and TLS framing, so the two can be checked against each other. The 'framing' rows hold the records that carry no identity material: `ChangeCipherSpec`, the one-byte application probe of Section 3, and the post-handshake records. Byte counts in this table include the 5-byte TLS record header; the key-exchange totals quoted in the text are payload-only and are therefore 10 bytes smaller for the two hello messages, which is the 2,530 of Table 4.

Direction	Message	Bytes	Contents
client→server	`ClientHello`	1,357	of which key share 1,192
	`Certificate`	5,549	DER 5,514 + 35 framing
	`CertificateVerify`	3,339	signature 3,309 + 30 framing
	`Finished`	74	
	framing	53	CCS 6, probe 23, close 24
server→client	`ServerHello`	1,183	of which key share 1,096
	`EncryptedExtensions`	28	
	`CertificateRequest`	107	
	`Certificate`	5,549	as above
	`CertificateVerify`	3,339	as above
	`Finished`	74	
	framing	188	CCS 6, tickets 158, close 24
Total, both directions		20,840	

Taking the two hello messages as the key-exchange term, consistently with Table 4, they total 2,530 bytes or 12.1 % of the handshake; of those, 2,288 carry the actual key material (1,184 bytes of ML-KEM public key and 1,088 of ciphertext, plus extension headers) and the rest is TLS machinery. The identity material is 18,310 bytes.

Every row of the table can be checked against the standards rather than against another measurement, and the largest row is the one worth checking. The 5,549-byte `Certificate` is 35 bytes of TLS framing — 5 of record header, 16 of AEAD tag, 4 of handshake header, 1 of content type, 1 of context length, 3 of list length, 3 of certificate length and 2 of extensions length — plus 5,514 bytes of DER; that DER is in turn 1,952 of ML-DSA-65 public key, 3,309

of signature and 253 of X.509 framing (version, serial, algorithm identifiers, names, validity, extensions and the signature wrapper). The same three numbers appear in every certificate in this paper, which is why a chain of n certificates costs what it does in Section 5.3. The mechanism is therefore visible per message as well as in aggregate: it is not that ML-KEM is large, it is that there are two certificates and two signatures, and each of the four is larger than the entire key exchange.

The two ratios quoted in the introduction differ for a reason worth naming. The 7.8 uses the standard object sizes alone, 17,646 against 2,272; the 7.2 uses what the wire carries, 18,310 against 2,530. Framing raises both terms but the smaller one faster — 2,272 to 2,530 is 11.4 %, 17,646 to 18,310 is 3.8 % — so the ratio a packet shows is the lower of the two. Neither is an error; a paper that quoted only the higher one would be quoting a number no link can produce.

Mutual authentication turns out to be almost free, which is a result rather than an oversight. Every number above authenticates both ends. A reader may reasonably ask what authenticating one direction would save, since an uplink that dials a known server is a common shape and MAVLink signing is sometimes offered as the aircraft-side check. We measured it (`results/oneway.csv`): omitting the client certificate removes 8,995 of the 20,840 bytes, 43 %, and 0.10 s of the 11.28 s. The bytes fall by nearly half and the time barely moves, because completion time is set by the longest flight in either direction rather than by the total. The server's flight is 10,468 bytes with a client certificate on the wire and 10,361 without — the certificate the server sends does not shrink — so the client's certificate rides in parallel with a wait that was going to happen anyway. Mutual authentication starts to cost something only when the client's flight would exceed the server's, and at these sizes it does not.

Two consequences follow. The first is that the mutual-authentication numbers throughout this paper are the conservative ones, and an operator cannot buy airtime back by authenticating one direction only. The second is a change in the split: with one direction authenticated, the identity share falls from 87.9 to 78.6 %, because the 2,530 bytes of key exchange are not halved along with the certificates. That is a change in the denominator, not in the remedy — identity still dominates, and the three remedies retain their order.

4.3. The Certificate Parameter Sweep

ML-DSA has three parameter sets, and because they differ only in security level they provide a controlled probe of how bytes convert into time. Table 8 reports all four certificate algorithms at a fixed key exchange.

Table 8. Handshake size and completion time for mutual authentication, ML-KEM-768 key exchange, varying certificate signature algorithm (9.6 kbps per direction, 250 ms one-way). Times are single runs; the fifth column is the sample standard deviation over five repeats of the same cell on the same link (`results/sweep_repeat.csv`), whose mean lies within 0.03 s of the single-run time in the same row, and the final column is the byte difference between adjacent rows divided by the time difference. The classical row is the only one whose byte count moves between batches, for the reason Section 4.1 gives.

Certificate	Bytes	9.6 kbps (s)	64 kbps (s)	sd, 5 runs	Marginal (B/s)
ECDSA-P256	4,114	3.81	0.58	0.024	–
ML-DSA-44	16,004	9.02	1.74	0.018	2,282
ML-DSA-65	20,840	11.27	2.08	0.012	2,150
ML-DSA-87	27,392	14.27	2.45	0.013	2,183

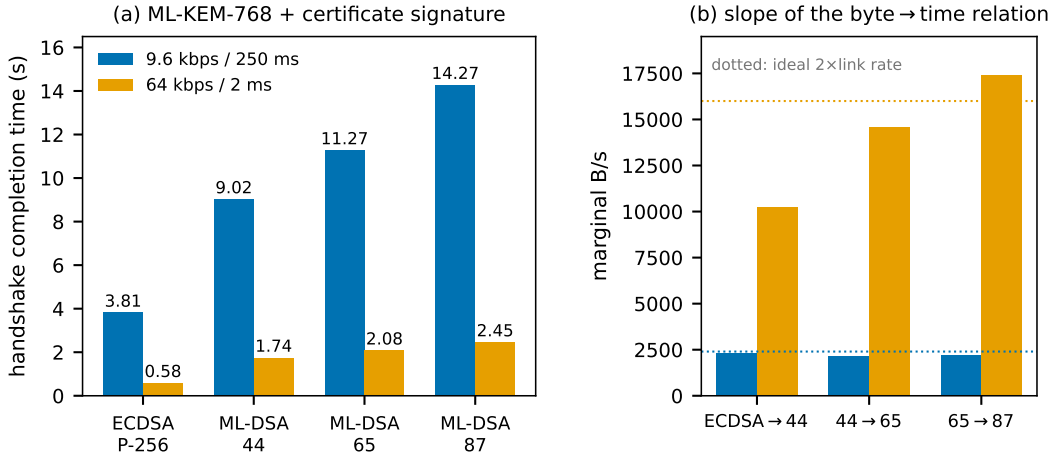


Figure 3. (a) Completion time for the sweep of Table 8 at two link rates. (b) The slope of the same sweep, computed as the byte difference between adjacent certificate algorithms divided by the time difference. At 9.6 kbps the slope is stable at 2,150–2,282 B/s, which is 90–95 % of the ideal 2,400 B/s obtained if both directions serialise in parallel at the link rate. This stability is what licenses the extrapolation in Figure 7; at 64 kbps the same slope scatters by a factor of 1.7, so we do not extrapolate from that rate.

Figure 3(b) is the reason this sweep is worth reporting rather than merely listing. The marginal throughput at 9.6 kbps is 2,282, 2,150 and 2,183 B/s for the three steps, a spread of 1.5 %. The ideal value, if both directions serialise in parallel at the link rate, is $2 \times 1{,}200 = 2{,}400$ B/s; the measured values are 90–95 % of it, the shortfall being TLS and TCP framing. At 64 kbps the same computation gives 10,236, 14,572 and 17,425 B/s, a spread of 70 %, because at that rate the fixed protocol overheads are comparable to the transfer itself.

The practical consequence is a conversion rule with a stated domain of validity. On a 9.6 kbps link with 250 ms propagation, adding a byte to a handshake costs $1/2{,}200 \approx 0.45$ ms. That rule can be used to predict configurations we cannot measure directly, which is what Section 4.8 does; it should not be used at rates where the fixed overheads dominate.

4.4. How the Penalty Moves with Link Rate

Every measurement so far is at 9.6 kbps per direction. A byte-count penalty should shrink as a link speeds up, so a reader is entitled to ask whether the 5.66× of Table 11 is an artefact of having chosen a slow link. We tested that by varying one thing: rate from 2.4 to 38.4 kbps in five steps, with the 250 ms one-way delay held fixed, so that every point describes the same geostationary link at a different transponder bandwidth. That is how satellite C2 capacity is actually bought, so the sweep corresponds to a procurement decision an operator will face. Two runs per cell.

Table 9. Completion time against link rate at a fixed 250 ms one-way delay. Cells are mean handshake bytes / mean completion time in seconds over two runs. Two is fewer than elsewhere in the paper, and the justification is the measured spread rather than convenience: across the fifteen cells the largest within-cell difference is 1.9 % and the median 0.3 %, both far below the 2.8× range the table exists to show; a standard deviation from two runs would carry no information, so the range is quoted in its place. "PQC" is ML-KEM-768+ML-DSA-65, "pre-shared" is ML-KEM-768 with an external PSK. Ratios are formed within each row, so they are unaffected by the inter-run byte variation of Section 4.1.

Rate (kbps)	Classical	PQC	Pre-shared key	PQC / classical	PSK / classical
2.4	1,913 / 4.68	20,658 / 40.43	2,902 / 10.09	8.64×	2.16×
4.8	1,915 / 2.88	20,840 / 20.99	2,902 / 5.56	7.29×	1.93×
9.6	1,913 / 1.98	20,840 / 11.27	2,902 / 3.37	5.70×	1.70×
19.2	1,912 / 1.56	20,840 / 6.41	2,902 / 2.18	4.11×	1.40×
38.4	1,913 / 1.30	20,840 / 3.98	2,902 / 1.61	3.07×	1.24×

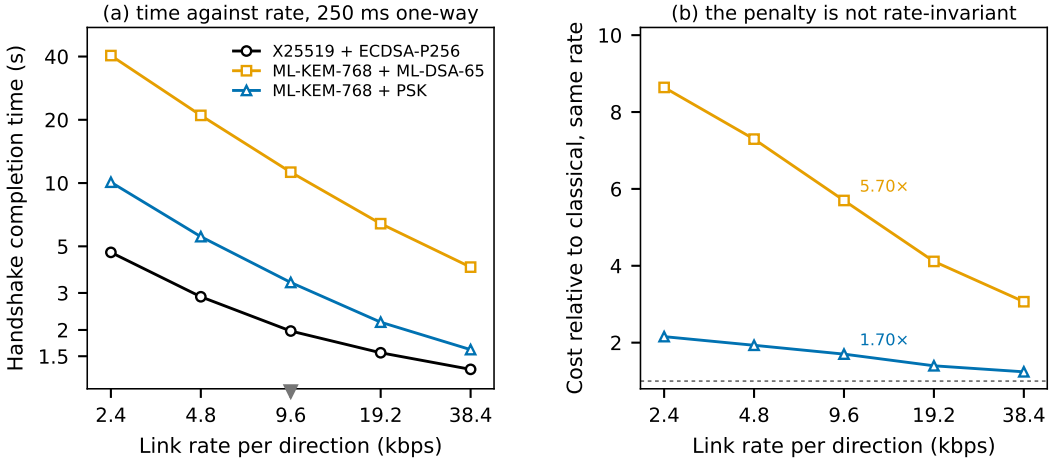


Figure 4. Rate sweep at a fixed 250 ms one-way delay. (a) Completion time against rate on logarithmic axes; the small triangle below the axis marks the 9.6 kbps DO-362 operating point used everywhere else in this paper. A handshake bound purely by serialisation would follow a line of slope −1; the measured lines are steeper at the low end and flatten at the high end, which is a fixed cost being overtaken by a rate-proportional one. (b) The same data expressed as a cost relative to the classical baseline at the same rate. The certificate penalty is not rate-invariant: it falls from 8.64× to 3.07× across the sweep. The pre-shared-key penalty falls more gently, from 2.16× to 1.24×, because it has removed most of the bytes that scale with rate. The annotated 9.6 kbps values (5.70× and 1.70×) belong to this sweep's own baseline of 1.98 s and are therefore 0.7 % and 1.8 % above the 5.66× and 1.67× of Table 11, which comes from a different batch; see the batch rule in Section 4.1.

Figure 4 plots the sweep, both as absolute completion time and as the same data expressed relative to the classical baseline at each rate. The figure is its own batch, so its ratios sit 0.7 % and 1.8 % above the 5.66× and 1.67× of Table 11; the batches differ only in the ECDSA baseline, for the reason Section 4.1 gives.

The penalty is not a constant, and the earlier draft of this paper said it was. It falls monotonically from 8.64× at 2.4 kbps to 3.07× at 38.4 kbps. The 5.66× that organises most of the discussion above is therefore the value *at one operating point*, not a property of the handshake. Reporting it without the sweep would have overstated the migration cost at high rate and understated it at low rate, and the understatement matters more, because it is the narrow links that need the migration most.

Why the ratio moves while the byte difference does not. At 2.4 kbps the row reads 20,658 bytes against a 1,913-byte classical baseline, a gap of 18,745; the four faster rows read

20,840 against baselines of 1,912 to 1,915, gaps of 18,925 to 18,928. So the gap is constant to within four bytes above 2.4 kbps and 183 bytes lower at the slowest rate, and both halves of that 183 have the same cause, the window-boundary effect described at the end of this subsection: the certificate handshake counts 182 bytes fewer (20,658 against 20,840) and the classical baseline one byte more (1,913 against 1,912 at 19.2 kbps). Neither changes a ratio in the third significant figure. What changes is how much of the completion time is serialisation. At 38.4 kbps the classical handshake takes 1.30 s, of which serialising its larger direction (1,048 bytes) accounts for 0.22 s; at 2.4 kbps it takes 4.68 s against 3.49 s. The ratio of measured to serialisation-only time therefore rises from 1.34 to 6.0 as the link speeds up, and the fixed cost — two propagation legs plus the handshake round trips — is diluted by a denominator that has stopped growing. The absolute gap tracks serialisation instead: it widens thirteen-fold, from 2.68 s at 38.4 kbps to 35.75 s at 2.4 kbps, while the bytes responsible for it do not move at all.

The pre-shared-key option is far less rate-sensitive. Its penalty falls only from 2.16× to 1.24×, because 87.9 % of the certificate handshake's bytes are gone and with them most of the term that scales with rate. This strengthens rather than weakens the design rule of Section 4.6: the architecture choice is worth 2.5× in time at 38.4 kbps and 4.0× at 2.4 kbps, against the 1.25× that the largest signature-algorithm change available (ML-DSA-65 down to ML-DSA-44) delivers at 9.6 kbps — and the certificate rows of Table 9 show the penalty growing, not shrinking, as the link narrows, so the gap between the two kinds of remedy widens across the measured range. Below 2.4 kbps we have no measurement, and the trend is an extrapolation there.

One number needs a caveat. The 2.4 kbps row counts 20,658 bytes rather than 20,840, 0.9 % fewer. The record decomposition at that rate is identical to the one at 9.6 kbps (same record lengths, same order), so the difference is a counting artefact at the measurement window boundary rather than a differently framed handshake. At 2.4 kbps those 182 bytes are worth 0.61 s, which is 1.5 % of the 40.43 s in that row; the ratio we report for the row would move in the third significant figure and we quote it to two decimals accordingly.

4.5. How the Penalty Moves with Link Delay

The sweep above varies bandwidth at a fixed 250 ms delay, which leaves a second objection open: apart from that sweep, every number in this paper describes one geostationary link. An operator flying a low Earth orbit constellation will want to know whether the 5.66× survives a 20–60 ms path, and whether "bandwidth is the binding constraint" survives a link whose latency is small.

One caveat belongs before the numbers rather than in the caption. The emulator charges its configured delay once per forwarded chunk rather than once per direction switch, so the *absolute* slope of time against delay is a property of the emulator, not of the link, and cannot be compared with a real satellite budget. What is transport-independent is the ratio between two configurations measured on the same emulator, and that is all this section reads from the table.

We varied the one-way delay and held everything else fixed: 9.6 kbps per direction, three configurations, four delays, three runs per cell, one batch. Zero delay isolates serialisation; 40 ms is a low Earth orbit pass; 250 ms is geostationary; 600 ms is geostationary with a second hop through a ground station, which is the topology an operator with a remote control centre actually builds. Table 10 gives the result.

Table 10. Completion time against one-way propagation delay at a fixed 9.6 kbps per direction. Cells are means over three runs; the largest standard deviation in the table is 0.014 s. "PQC" is ML-KEM-768+ML-DSA-65, "pre-shared" is ML-KEM-768 with an external PSK. Ratios are formed within each row. This is a third batch, so its 250 ms row reads 5.76× where Table 11 reads 5.66× and Table 9 5.70×; the byte counts are identical and the difference is the classical baseline of Section 4.1. The slope these rows imply is a property of how the emulator charges delay, not of the handshake; only the ratio between the two slopes is read as a link property (Section 4.5).

Delay (ms)	Link	Classical (s)	PQC (s)	Pre-shared key (s)	PQC / classical	PSK / classical
0	control, no propagation	1.47	10.27	2.44	7.00×	1.66×
40	low Earth orbit	1.55	10.43	2.62	6.71×	1.69×
250	geostationary	1.96	11.27	3.31	5.76×	1.69×
600	geostationary, two hops	2.68	12.67	4.00	4.72×	1.49×

Two things move in opposite directions, and until now this paper has been quoting the one that flatters the technology. Completion time is linear in delay to within 14 ms across the four rows, with a slope of 4.00 s per second of one-way delay for the certificate handshake, 2.03 for the classical baseline and 2.59 for the pre-shared-key one. The certificate configuration therefore places twice as much of its work behind a propagation delay as the classical one, and the absolute gap between them *widens*, from 8.80 s at zero delay to 9.99 s at 600 ms. Latency does not rescue the certificate handshake; it adds to it.

The ratio, however, falls from 7.00× to 4.72×, because the classical baseline nearly doubles over the same interval (1.83×) while the post-quantum total grows by 1.23×. The byte gap is constant at every delay; what changes is the common additive term that latency contributes, which inflates a smaller denominator proportionally more. This is the same mechanism as in the rate sweep, with latency in place of rate, and it produces the same kind of movement in the ratio and the same stability in the bytes.

The consequence is that the headline figure is conservative. The ratio falls monotonically with delay, so for any link with no more than 250 ms of one-way propagation — low Earth orbit, medium Earth orbit, or a single geostationary hop — the certificate penalty is at least the 5.66× of Table 11: 6.71× at 40 ms and 7.00× at zero delay. Only the two-hop path buys the ratio down to 4.72×, and it pays for that with an absolute penalty 0.7 s larger than the geostationary one. A reader who deploys over low Earth orbit should expect the ratio of Section 4.1, not a smaller one, and the absolute penalty of 8.9 s there is within 0.5 s of the geostationary figure. The claim that narrow links are where the certificate penalty bites is thus reinforced rather than qualified: the links with the least latency are the ones where it is largest relative to the baseline they replace.

Three caveats bound this table. The slope counts chunk-level delay payments, because the proxy holds each forwarded chunk for one one-way delay (Section 3.2); the absolute value 4.00 is therefore a property of the emulator, and only the ratio of slopes — twice the classical figure — is interpretable as a property of the handshake. Zero delay is a control and not a link: no radio path has none, and the row exists to separate serialisation from propagation. And the table is a batch of its own, as the caption states.

4.6. Pre-Provisioned Trust

The sweep suggests that shrinking the certificate is expensive per byte saved. The alternative is to not send a certificate at all. A TLS 1.3 external pre-shared key authenticates both endpoints and establishes fresh key material without any certificate, any `CertificateVerify` and any trust anchor on the link.

Table 11 reports the comparison; its classical baseline is the 1,915-byte, 1.99 s measurement of the same batch as Table 5, which is why the ratio here is 5.66× and the one in

Figure 4 is 5.70×. The pre-shared-key handshake carries 2,902 bytes and completes in 3.32 s, against 20,840 bytes and 11.29 s for certificate authentication: 7.2× less traffic and 3.4× less time. It retains ML-KEM-768 for key exchange, because TLS 1.3 pre-shared-key mode uses `psk_dhe_ke` by default and therefore keeps the post-quantum contribution to forward secrecy. The records confirm the mode: the client's `ClientHello` grows by exactly 50 bytes, which is the `pre_shared_key` extension, while the server still returns a 1,088-byte ML-KEM ciphertext. The 17,938 bytes saved are therefore identity material, not key exchange. What it gives up is identity: the peers are authenticated by possession of the pre-shared key, not by a certificate chain.

Two things follow that the byte count does not show, and both cut against reading the 7.2× as a free saving. The first is that the risk is concentrated rather than created: a pre-shared key authenticates every session that holds it, so its compromise is the compromise of the whole fleet at once, whereas one aircraft's signing key opens one aircraft. The radius follows from the topology rather than from the crypto: in the twenty-aircraft, three-station fleet of Section 5.4 a captured aircraft key opens three links for the weeks it takes to re-key them, and a group key used instead of pairwise keys opens all sixty endpoints for the same window. The second is that the certificate has a revocation channel and the pre-shared key does not. A certificate can be withdrawn and the peer can be told; a key can only be replaced, and replacing it means reaching every aircraft. The bytes of that channel are not in any table here — a CRL or an OCSP response on the same 9.6 kbps link is itself hundreds to thousands of bytes — so the comparison is of airtime per handshake, not of lifecycle cost, and the two are not commensurable on the evidence collected.

Table 11. Certificate authentication versus pre-provisioned trust, mutual authentication, 9.6 kbps per direction, 250 ms one-way. The classical baseline is included for scale. The pre-shared-key configuration keeps ML-KEM-768 for key exchange and drops only the identity material. Its 2,902 bytes comprise 330 bytes of hello common to every configuration, 56 bytes of pre-shared-key extension and 2,200 bytes of ML-KEM-768 exchange, so the 1,088-byte ciphertext is still on the wire; that is what preserves forward secrecy, and it is why the saving is 3.4× rather than the 30× that also dropping the post-quantum exchange would give. The classical baseline is 1.99 s from the same batch, so the 5.66× here is formed against that; Figure 4 quotes 5.70× from the rate sweep's own 1.98 s baseline.

Configuration	Handshake bytes	Time (s)	Relative to classical
X25519 + ECDSA-P256 (classical)	1,915	1.99	1.00×
ML-KEM-768 + ML-DSA-65	20,840	11.29	5.66×
ML-KEM-768 + pre-shared key	2,902	3.32	1.67×

https://doi.org/10.3390/------

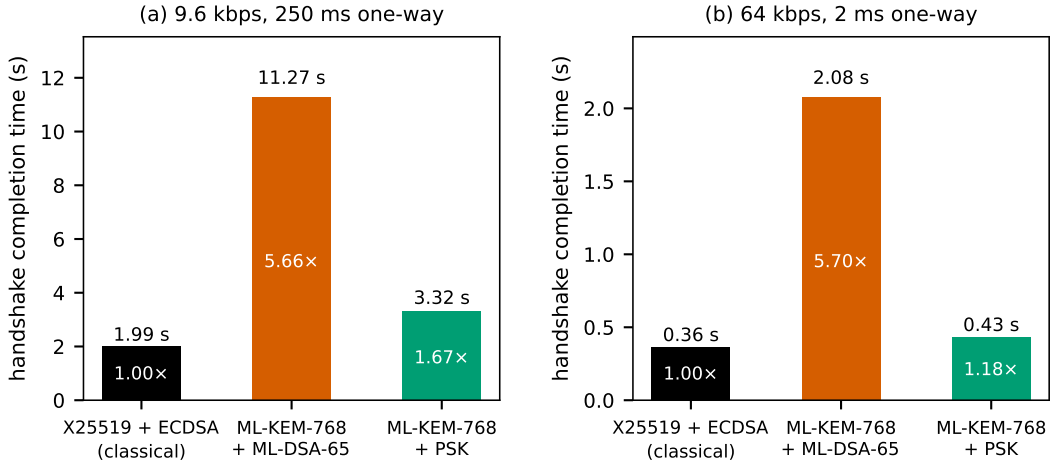


Figure 5. The same three configurations on two different links: a 250 ms, 9.6 kbps beyond-line-of-sight link and a 2 ms, 64 kbps line-of-sight one. The certificate handshake costs 5.66× the classical baseline on the first and 5.70× on the second, the pre-shared-key handshake 1.67× and 1.18×. The near-identity of the certificate penalty across two very different links is a coincidence of two opposing effects, a faster link shrinking the serialisation term but also the fixed term; Section 4.4 isolates them by holding the delay fixed.

Figure 5 repeats the comparison on the line-of-sight link of Section 2: 64 kbps and 2 ms. The certificate penalty is almost unchanged, 5.70× against 5.66×, but that agreement should not be read as the penalty being rate-invariant, because the two links differ in delay as well as in rate and the two effects happen to cancel. Section 4.4 separates them by holding the delay fixed, and finds the penalty ranging from 3.07× to 8.64×. What does transfer between links is the ordering: the pre-shared-key penalty falls from 1.67× to 1.18×, because on the faster link the handshake is short enough that fixed protocol overheads rather than payload dominate.

4.7. Link Loss: What It Changes and What It Does Not

Every measurement above is on a clean link, which is the least favourable case for our own argument: loss forces retransmission, a certificate is 5,500 bytes and therefore more segments that can be lost, so the natural prediction is that loss amplifies the certificate penalty. We tested that prediction and it is only half right.

A TCP proxy cannot drop a byte without corrupting the stream, and DTLS — which could drop datagrams honestly — is unavailable in this OpenSSL build, which offers only DTLS 1.0 and 1.2 and in any case failed to complete even a classical handshake in our environment. We therefore inject loss as a retransmission timeout: with probability p a chunk is held for one RTO of 1.2 s instead of being delivered, which reproduces the time-axis effect of a lost segment while leaving the byte stream intact. Loss is counted per 1,460-byte segment rather than per proxy chunk, because counting per chunk makes a 20,840-byte handshake suffer the same loss count as a 1,912-byte one — an earlier version of this experiment did exactly that and concluded, wrongly, that loss does not interact with handshake size.

Table 12. Effect of link loss at 9.6 kbps and 250 ms, eight runs per cell, mutual authentication. "Losses" is the mean number of injected retransmission timeouts actually suffered, measured, not assumed; it scales with the segment count rather than with p alone. Times are in seconds.

Configuration	Bytes	Losses	Clean mean	p=0.20 mean	ΔT (s)	ΔT (%)
X25519 + ECDSA-P256	1,912	1.0	1.96	2.41	+0.45	+22.8
ML-KEM-768 pre-shared key	2,902	1.0	3.31	3.75	+0.44	+13.4
ML-KEM-768 + ML-DSA-44	16,004	2.6	9.01	10.35	+1.34	+14.9
ML-KEM-768 + ML-DSA-65	20,840	3.4	11.28	12.61	+1.34	+11.9

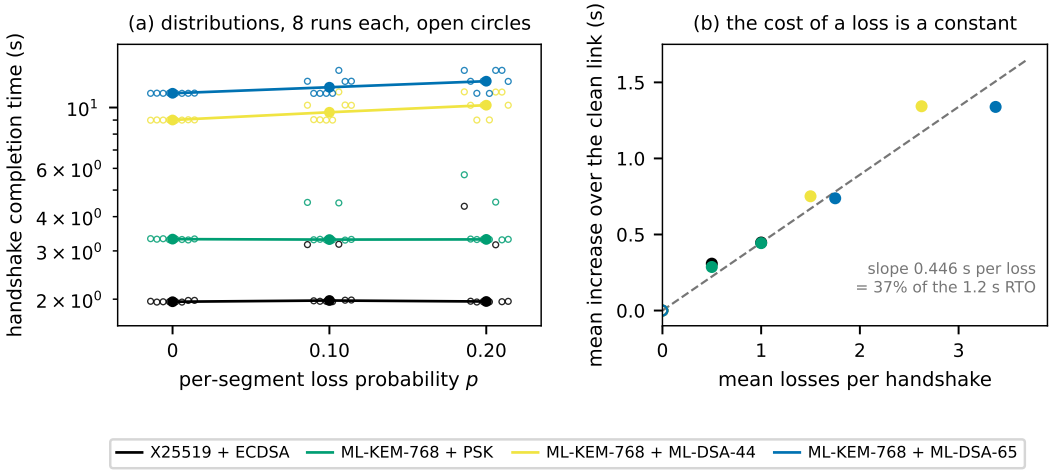


Figure 6. Link loss. (a) Completion time against per-segment loss probability; each open circle is one of eight runs, the filled circles and lines are the medians. The medians barely move while the spread grows, which is why a single run per cell would be misleading. (b) The mean increase over the clean link, for all twelve configuration-by-loss-rate points, against the number of injected retransmission timeouts actually suffered. The points fall on a line through the origin of slope 0.446 s per loss — 37 % of the 1.2 s timeout, the rest being absorbed by the reverse direction. Since losses in turn scale with segment count, this is the mechanism by which a larger handshake pays more: not a larger per-loss cost, but more losses.

Three things follow from Table 12 and Figure 6.

The cost of a loss is a constant, and it is not the whole RTO. Across all twelve configuration-by-loss-rate points, the mean increase in completion time is 0.446 s per injected loss, against an injected 1.2 s: 37 % of the timeout reaches the critical path, because a loss that lands while the peer is still transmitting in the other direction is partly absorbed. Figure 6(b) plots every point against the losses actually suffered and they collapse onto that single line through the origin, to within 0.2 s. The penalty is therefore a property of the link, not of the handshake.

What varies between handshake designs is how many losses they incur. A 1,912-byte handshake spans one to two segments and suffers one loss at $p = 0.20$; a 20,840-byte handshake spans sixteen — eight in each direction, because the proxy charges every forwarded chunk at least one segment and rounds a 4,096-byte read up to three — and suffers 3.4, against the 3.2 that sixteen segments at $p = 0.20$ predict. Multiplying by 0.446 s gives the observed +0.45 and +1.34 s. Loss does amplify the certificate penalty, but only through the segment count, and the amplification is modest: 1.34 s against the 9.3 s that the identity material costs in the first place.

The ranking does not change, and the gap between the two architectures is stable. Comparing like with like — every configuration at the same loss rate — ML-KEM-768+ML-DSA-65 costs 5.23× the classical baseline at $p = 0.20$, against 5.75× at $p = 0$ in the same

sweep, and the pre-shared-key case costs 1.56× against 1.69× at $p = 0$. The $p = 0$ figures come from the loss sweep's own baseline and therefore differ slightly from the 5.66× and 1.67× of Table 11; see the batch rule in Section 4.1. Both ratios fall rather than rise, because the classical handshake is small enough that a fixed 0.45 s penalty is a larger fraction of it; a first version of this comparison divided the lossy post-quantum figure by the clean classical one and produced 6.43×, which is an artefact of mixing conditions. What carries the argument is the ratio between the two post-quantum options, and that is essentially unchanged: 3.41× on a clean link, 3.36× at $p = 0.20$. Loss widens the absolute gap from 9.31 to 10.20 s but buys back none of what pre-provisioning saves. On a link losing one segment in five, pre-shared keys remain 3.4× cheaper in time and 7.2× in bytes.

One observation cuts the other way and is worth stating because it constrains how these numbers should be quoted. Loss does not shift the median much — the pre-shared-key median is unchanged at 3.31 s between $p = 0$ and $p = 0.20$, and the ML-DSA-65 median moves from 11.27 to 12.47 s — but it moves the mean, and it moves the maximum more. The classical baseline at $p = 0.20$ has a median of 1.96 s and a worst observed run of 4.36 s, a spread of 2.2× from the same configuration. A retransmission does not merely add latency; it triggers congestion control, and the resulting stall can exceed one RTO. Quoting a mean, or a single run, conceals this. It is the reason Section 4.9 reports distributions rather than point estimates, and the reason a link budget for an atomic transaction should be set from a high quantile.

The model has two limitations, and they can be netted off rather than merely confessed. The per-segment loss draw is binomial and independent, so it cannot represent the bursty fading that dominates a real narrowband radio link; and charging one RTO per lost segment is an upper bound, since a real TCP retransmits several losses within one timeout using SACK. Take a Gilbert–Elliott channel whose bad state loses every segment, with stationary loss rate p and mean burst length L segments. A handshake of n segments then meets about np lost segments but only np/L distinct recovery episodes, and it is episodes, not segments, that a SACK stack charges. Two assumptions sit in that count and are worth naming: L is a length in segments, and it is the *total* loss in the bad state that makes the count exactly np/L — a bad state dropping only a fraction p_{bad} of its segments would multiply the count by $1/p_{\text{bad}}$, so $1/L$ is the optimistic end of the family rather than a general scaling law. The mean penalty therefore scales as $1/L$: at $p = 0.20$ the measured +1.34 s becomes +0.48 s at $L = 3$ and +0.14 s at $L = 10$ (`harness/burst_bound.py`). Since both the losses and the episodes are proportional to n, the *ratio* between two configurations is invariant to L — which is a stronger statement than Section 4.7 makes, because it says the ordering survives burstiness by construction rather than by coincidence. The direction that does not survive is the tail: a burst of L segments is $L \times 1.22$ s of dead air at 9.6 kbps, so $L = 10$ stalls for 12.2 s, while the largest stall this model can produce is one RTO. The means in Table 12 are therefore an upper bound and the worst case is a lower bound, which is the direction a link budget should err in but not one to quote without saying so.

The loss rates in the table are a stress case, not an operating point. Ten and twenty per cent per segment are one to two orders of magnitude above a link that is engineered at all, and a reader is entitled to ask what happens where links actually run. We therefore repeated the three configurations at 0.1, 0.5 and 1 % per segment, five runs per cell, in a batch with its own 0 % baseline (`results/loss_low.csv`). The certificate handshake moves by −0.009, −0.002 and −0.005 s — a shift that is not resolvable in either direction, and whose sign is not even the expected one. Across all nine cells of the batch the deltas span −0.009 to +0.013 s against standard deviations of 0.008–0.019 s measured in the same cells. The slope of Figure 6(b) predicts 0.16 injected losses at 1 % and therefore 0.07 s, which is inside the noise a clean link already has. Two conclusions follow, and we keep both: the

rows of Table 12 describe an upper regime that bounds what loss can do rather than what it does, and the model is self-consistent at the low end, in that a prediction of 0.07 s is not falsified by a measurement whose magnitude is 0.005 s. What the low-loss batch cannot show is that loss is harmless on a real radio link, because a link loses segments while this model loses chunks; the honest statement is that between 0 and 1 % the effect is smaller than the run-to-run spread at five runs per cell — not that it is zero — and above 5 % it is not.

4.8. What Compact Signatures Can and Cannot Buy

The remaining question is whether a smaller signature scheme would do as well as removing the certificate. FN-DSA (Falcon) offers 666-byte signatures and 897-byte public keys, the smallest of any scheme on a standardisation track, and it is the obvious candidate.

OpenSSL 3.5 does not implement Falcon, so we cannot measure it. We therefore extrapolate along the slope measured in Section 4.3, and we label the result a prediction throughout. An FN-DSA-512 certificate replaces 1,952 + 3,309 bytes of key and signature material with 897 + 666; doubled for mutual authentication the handshake loses 7,396 bytes, from 20,840 to 13,444. At the 2,183 B/s marginal that Table 8 measures between ML-DSA-87 and ML-DSA-65, that is a saving of 3.4 s, giving 7.88 s.

The slope is not one number, and the prediction should not be quoted as if it were. The three marginals of Table 8 are 2,150, 2,183 and 2,282 B/s; across that range the same 7,396-byte reduction gives 7.83 to 8.03 s, so the gain over ML-DSA-44 is 11 to 13 % rather than a point value. The conclusion below does not depend on which end is used: no point in the range closes the gap to removing the certificate.

Table 13. The byte-cost ladder, measured points and one prediction. The ML-KEM-768 + FN-DSA-512 row is *predicted*, not measured: OpenSSL 3.5 has no Falcon implementation, so its byte count follows from the object sizes and its time from the slope of Section 4.3. All other rows are measurements.

Configuration	Handshake bytes	Time (s)	Source
ML-KEM-768 + ML-DSA-87	27,392	14.27	measured
ML-KEM-768 + ML-DSA-65	20,840	11.27	measured
ML-KEM-768 + FN-DSA-512	13,444	7.88	*predicted*
ML-KEM-768 + ML-DSA-44	16,004	9.02	measured
ML-KEM-768 + pre-shared key	2,902	3.32	measured
X25519 + ECDSA-P256	1,915	1.99	measured

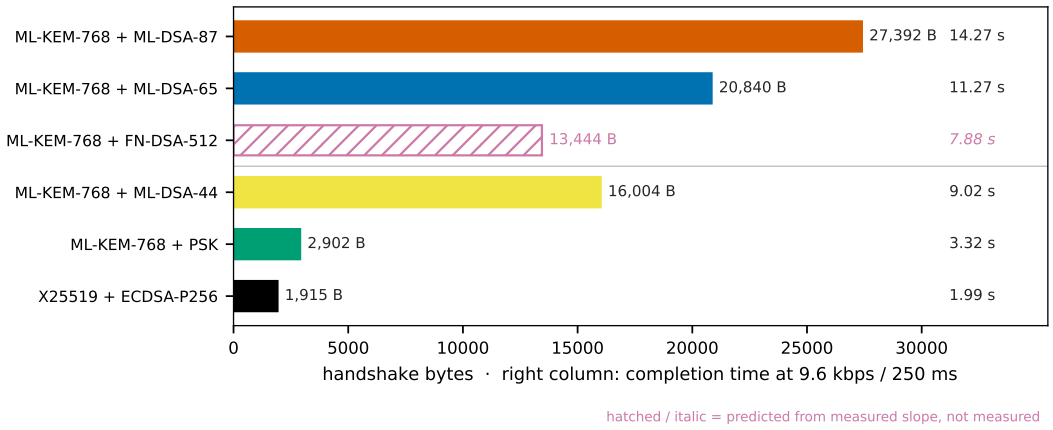


Figure 7. The byte-cost ladder. The hatched bar is the FN-DSA-512 prediction; every other bar is measured, and the figure's own annotation repeats the distinction. FN-DSA is the smallest signature scheme available and it does buy something — 7.88 s against 9.02 s for ML-DSA-44, or 7.83–8.03 s across the range of measured marginals — but the gain is 12.6 %, and it is smaller than the gain from dropping one parameter set. Removing the certificate, at the bottom of the figure, is worth 5.7 s.

The prediction is worth stating plainly because it contradicts the intuition that smaller signatures are better. FN-DSA-512 does buy something: 7.88 s against 9.02 s for the already standardised ML-DSA-44, a gain of 12.6 %. But the gain is less than one step on the ML-DSA parameter ladder, and it is bought with three costs that do not appear in the byte count. FN-DSA signing on Cortex-M4 takes 19.6 million cycles in Pornin's constant-time implementation, about 117 ms at 168 MHz, against roughly 50 ms for ML-DSA-65 [32] — a factor of 2.3 in the direction of the computation we argued at the outset is not the bottleneck, but which becomes relevant if the design is otherwise optimised. The expanded private key needs 40–57 kB of RAM, which is a substantial fraction of the 192–512 kB available on a typical flight controller. And FIPS 206 has not been finalised, so a design that depends on FN-DSA depends on a moving target.

For completeness: SLH-DSA is not a candidate here. Its smallest parameter set carries 7,856-byte signatures (Table 1, from FIPS 205), 3.2 times the ML-DSA-44 signature, and its signing time is measured in hundreds of milliseconds on a microcontroller [8], against well under a millisecond for the lattice schemes. Both costs are worse at every parameter set, and it is the right choice only where hash-based assumptions are required, which is not the case for a session handshake.

4.9. Repeatability

The sweep in Table 8 reports single runs, because a 9.6 kbps handshake takes long enough that repeating the full matrix is expensive. To establish that the numbers are not noise, we repeated the three configurations that carry the argument five times each (Table 14).

Table 14. Repeatability, five runs per configuration at 9.6 kbps per direction and 250 ms one-way, one batch. Byte counts are identical across runs; time varies by less than 1 %. The individual runs are kept in `results/repeatability_raw.csv`, and two further batches of the same kind are discussed below.

Configuration	Bytes	Mean time (s)	SD (s)	Range (s)
X25519 + ECDSA-P256	1,916	1.989	0.013	1.967–2.002
ML-KEM-768 + ML-DSA-65	20,840	11.276	0.011	11.262–11.290
ML-KEM-768 pre-shared key	2,902	3.314	0.014	3.298–3.327

The standard deviations are 0.65 %, 0.10 % and 0.42 % of the means, and the three means agree with the corresponding single runs in Table 8 to within 0.02 s. The differences the paper relies on — 11.28 against 3.31 s, or 9.02 against 11.27 s — are two to three orders of magnitude larger than the run-to-run variation, so the conclusions do not depend on measurement precision. The 1,916-byte classical figure here is one of the four baseline values catalogued in Section 4.1; the byte-level differences among them are DER encoding noise, not measurement error.

Two further batches of the same three configurations were run after the table was compiled — fifteen runs per configuration in total (`results/repeatability_raw*.csv`) — and they say something the single batch could not. *The means reproduce; the spreads do not.* Pooled over fifteen runs the means are 2.004, 11.285 and 3.340 s, within 1 % of the table, but the coefficients of variation rise to 4.6, 0.16 and 3.4 % because two of the forty-five runs stalled: one pre-shared-key run at 3.74 s against a 3.31 s median, and one classical run at 2.33 s against 1.96 s. Both are stalls of 0.4–0.5 s with no injected loss, which is the same order as the 0.446 s that a single retransmission costs in Section 4.7. A batch of five can miss such an event entirely, and the batch in Table 14 did; what protects the paper's conclusions is that the smallest effect it resolves, the 20 % between ML-DSA-65 and ML-DSA-44, is 125 times the pooled coefficient of the two configurations it compares (0.16 %) and 4.3 times the worst in the batch (4.6 %), which belongs to the classical baseline rather than to either of them. Both quantities are relative, which is what makes the comparison meaningful; they are not the same statistic, which is why both are given. That is a narrower margin than the first draft of this section claimed, and it is the honest one.

5. Discussion

5.1. Design Rules

The measurements reduce to five rules, in descending order of value.

Pre-provision trust if the operational model permits it, and only if you already distribute symmetric keys. This is worth 7.2× in bytes and 3.4× in time, and it is the only intervention that brings a post-quantum C2 handshake within 1.67× of its classical cost. It is also the only one that requires an architectural decision rather than a configuration change, and the precondition is not decorative: the saving is airtime, while the cost is a key-distribution mechanism the operator must already own. Section 5.2 states when the option is unavailable and Section 5.4 when it is available but not worth buying.

Do not spend effort on the key exchange. The difference between X25519 and ML-KEM-768 is 2,200 bytes; the difference between ML-KEM-768 and ML-KEM-1024 is a further 864. Both are small against the 16,726-byte identity term, and ML-KEM-768 is the right default for a C2 link: category 3, and the ciphertext fits comfortably in the objects the handshake already carries.

Choose the certificate parameter set for security, not for size. ML-DSA-44 saves 2.25 s against ML-DSA-65 at 9.6 kbps. For scale, one retransmission on the same link costs 0.45 s (Section 4.7), so the saving is worth about five losses: real, but the kind of margin that a bad link day absorbs. If the deployment requires category 5, ML-DSA-87 costs 3.00 s more than ML-DSA-65, also worth having measured, and a defensible price.

Do not build on FN-DSA for this application. The 12–13 % gain over ML-DSA-44 does not justify a draft standard, a 2.3× slower signature, and 40–57 kB of key state.

Authenticate both directions, because it is free. Removing the client certificate halves the byte count — 20,840 to 11,845 — and saves 0.10 s of 11.28 s, since the server's flight is the long one and the client's certificate travels underneath it (Section 4.1). Authenticating one direction only to save airtime therefore buys 43 % of the bytes and 0.9 % of the time. Spend the certificate at both ends.

5.2. Why Not Simply Use Symmetric Keys

The natural objection to the pre-shared-key recommendation is that it assumes an operational model many UAV operators do not have. It is worth being precise about where the recommendation applies, because the boundary is what determines whether the paper is useful to a given reader.

Pre-provisioned trust works when one organisation controls both endpoints and can distribute key material out of band: a fleet operator provisioning aircraft and ground stations, or a closed mission with a fixed set of participants. In that regime the recommendation is not merely available but obviously correct, and the classical baseline already relies on it — an operator running ECDSA today is relying on a certificate infrastructure they control.

It does not work when the aircraft must establish a session with an entity it was not provisioned against: a ground station operated by a different organisation, an air traffic service provider, a UTM service, or a relief aircraft from another fleet. Beyond-visual-line-of-sight operations in controlled airspace require exactly this kind of cross-organisational authentication, and there public-key cryptography is not optional. For that case the certificate measurements of Section 4.3 apply, and the honest reading is that certificate-based post-quantum C2 on a 9.6 kbps link costs about eleven seconds per handshake. Whether that is acceptable depends on how often the handshake happens, and session resumption is the obvious way to keep it rare. It is worth measuring rather than assuming, and the measurement corrects an earlier reading of our own: we had reported that OpenSSL 3.5.7 cannot resume a TLS 1.3 session at all, because `-sess_in` re-ran the full handshake. That diagnosis was wrong in its cause. `-sess_out` works; what was missing was a ticket to save. Every other measurement in this paper runs with `-no_ticket`, and a client that is never sent a `NewSessionTicket` has nothing to resume.

With tickets enabled, resumption is measurable, but only for server-authenticated handshakes. OpenSSL 3.5.7's `s_server` issues no ticket whenever client authentication is requested. The invocation, with `-verify 0` and `-Verify 1` substituted in turn for the last option pair, is

```
openssl s_server -accept 4433 -tls1_3 -groups MLKEM768
-cert leaf.pem -key leaf.key -CAfile leaf.pem -Verify 0
```

No variant sent a `NewSessionTicket`, and a ticket appears only when the request for a client certificate is dropped altogether. All three were run on 3.5.7 only: we did not test other patch releases, and the claim is scoped to the version measured. The reason is in the mode rather than in a bug: a resumed TLS 1.3 handshake authenticates with the PSK and runs no certificate authentication in either direction, so a server that requires a client certificate on every connection has nowhere to get one from. RFC 8446 [30] does not forbid the alternative — a server may bind the authenticated client identity to the ticket and resume on that basis, and other stacks do — so this is a behaviour of one server implementation, and one more reason to read the numbers here as stack-scoped (Section 5.6).

For the configuration in which it does work, measured on the same link:

- full server-authenticated handshake, 12,252 bytes and 11.26 s, of which 352 bytes are the ticket itself;
- resumed handshake, 3,335 bytes and 3.51 s: 0.27 of the bytes and 0.31 of the time.

Three things follow, and the first two cut against resumption as a substitute for the recommendation of Section 4.6.

Resumption is not cheaper than a bare pre-shared key. The external-PSK case of Table 11 costs 2,902 bytes and 3.31 s; resumption costs 433 bytes and 0.20 s *more*, because the

ticket is carried on the first handshake and again on the resumed one, while an operator holding a raw pre-shared key pays for neither.

For mutual authentication, resumption is not available in this stack at all. An operator in the cross-organisational regime — the regime that needs certificates in the first place — is running mutual authentication, and on OpenSSL 3.5.7 that means no ticket and no resumption. This is the paper's thesis showing up as a tool behaviour: the moment trust is pre-provisioned, certificates stop going on the wire, and a stack that must authenticate both directions by certificate cannot get there by resuming.

Where it does help, it helps a lot. A 3.5 s reconnect against an 11.3 s cold handshake is the difference between a session that can be re-established per task and one that cannot. The usable deployment is therefore server-authenticated transport with the aircraft's identity carried at the application layer, where a ticket can replace the handshake.

Three other alternatives are sometimes proposed and are worth dispatching. Enlarging classical parameters is strictly worse: no elliptic-curve parameter set resists Shor's algorithm. Quantum key distribution does not fit the platform — it needs a dedicated optical channel, a fixed line of sight to a trusted node, and an authenticated classical channel to bootstrap, which is the very problem it is supposed to solve. And accepting the risk is a legitimate decision rather than an oversight, provided it is made explicitly; Mosca's inequality [33] is the tool for making it explicit, and it requires an estimate of migration cost, which is what this paper supplies.

5.3. Chain Depth

Every certificate in the tables above is self-signed, so the chain depth is one. That is the right model for an operator who controls both endpoints, and it is the model under which an operator would run pre-provisioned trust anyway. It is not the model of a public key infrastructure, where a leaf is typically accompanied by at least one intermediate. We therefore built a two-level hierarchy — root signing an intermediate, intermediate signing the leaf, root trusted out of band and never sent — and measured it rather than extrapolating it.

The second link in the chain costs 11,248 bytes and 4.94 s at 9.6 kbps: a two-level handshake measures 32,088 bytes and 16.22 s against 20,840 and 11.28 s at depth one (Table 15). The rule is linear in depth, since each additional certificate is an independent list entry, so an operator with n intermediates should read every time in this paper as roughly $11.28 + 4.9 \times (n - 1)$ seconds at this link rate.

Table 15. Depth one against depth two, same algorithm (ML-DSA-65), same link (9.6 kbps, 250 ms), three repetitions each. Byte counts are identical across repetitions because every certificate is generated once and reused; the spread in time is 0.01 s.

Chain depth	Total bytes	Time (s)	vs depth one
One (self-signed leaf)	20,840	11.28	1.00×
Two (leaf + intermediate)	32,088	16.22	1.44×
Difference	+11,248	+4.94	

Where the 220 bytes beyond a naive estimate go. A first estimate treats the intermediate as a copy of the self-signed leaf and predicts $2 \times 5{,}514 = 11{,}028$ bytes; the measurement is 11,248, 220 more. The gap resolves to the byte, and the resolution is worth recording because it is a property of hierarchies rather than of this testbed. An ML-DSA-65 intermediate certificate encodes to 5,541 bytes, 27 more than the leaf, and each TLS list entry adds a three-byte length and a two-byte extensions field, so the two intermediates account for $2 \times (5{,}541 + 5) = 11{,}092$. The remaining 156 bytes are the leaf growing from 5,514 to

5,592 bytes when it stops being self-signed: a certificate that is verified against a parent rather than against itself carries `extendedKeyUsage` and `subjectAltName`, which the self-signed leaf does not need. Thus $2 \times 5{,}546 + 2 \times 78 = 11{,}248$, and the 220 is 2×27 for the intermediate's longer encoding, 2×5 for the list framing and 2×78 for the leaf's growth.

Why this is the more useful number. The per-direction figure can also be checked against the standard rather than against another paper: an ML-DSA-65 certificate carries a 1,952-byte public key and a 3,309-byte signature [3], leaving 253 bytes of X.509 framing, which with the 35 bytes of TLS framing is the 5,549-byte record of Table 3. Both routes agree, which is what makes the depth-two result safe to extrapolate further.

The direction of the error is the point. The absolute numbers in this paper are what an operator with a flat, self-signed hierarchy would see. An operator migrating a public hierarchy would see worse, and the degradation is linear in chain depth — which is a further argument for flattening the hierarchy during migration rather than afterwards. Note also that the 1.44× penalty for one extra link is larger than the 1.25× that the best signature-algorithm change buys (Section 4.3): adding a certificate to the chain costs more than every choice made inside it.

5.4. The Operational Cost of Pre-Provisioning

The recommendation to pre-provision is easy to state and harder to operate, and the difficulty is not cryptographic. Four problems dominate.

Provisioning. Key material has to reach the aircraft before the mission, through a channel that is itself trustworthy. Factory injection covers the aircraft but not the ground stations or the relief aircraft; field provisioning needs a mechanism and an audit trail, and the moment that mechanism exists it becomes an attack surface that did not exist when certificates were issued by a CA.

Rotation. A pre-shared key that is never rotated gives an adversary unlimited time against a single compromise. Rotation requires either connectivity at a known time — which a beyond-line-of-sight aircraft may not have — or a key hierarchy that supports derivation, which reintroduces some of the complexity the pre-shared key was meant to remove.

Revocation. A certificate can be revoked and the revocation distributed. A pre-shared key cannot be un-shared. If an aircraft is captured, its key is compromised until every peer it could talk to has been re-keyed, and the operator has to know which peers those were. The magnitude is worth stating, if only as an estimate: twenty aircraft against three ground stations is sixty pairwise keys, and the exposure lasts until each of them has been replaced, which for the aircraft means physical or out-of-band contact — a maintenance action of an hour each, so a sixty-hour campaign, and a window measured in weeks at the sortie rate used above rather than in hours. The certificate case is not free either, since a revoked chain still has to reach every station's revocation list, but it is a distribution problem rather than a fleet visit, and distribution completes in seconds over the same link that carries the handshake.

Scale. A pre-shared key is pairwise. A fleet of n aircraft and m ground stations needs up to nm keys, or a group key with the attendant problems of group rekeying. Certificates need one key per node plus a trust anchor.

The break-even, as arithmetic. A pre-shared key saves 7.97 s of link time per handshake against the depth-one certificate path, and 12.88 s once one intermediate is included. A small fleet — twenty aircraft flying twice a day, one handshake per sortie — completes 14,600 handshakes a year and therefore saves 32 hours of airtime, or 52 with an intermediate. Both are about one working week. The trade therefore does not turn on airtime at all: it turns on whether the operator already runs key distribution. An operator who signs

MAVLink v2 frames has that machinery already, because frame signing needs a shared secret at both ends (Section 2); for them the marginal cost of extending it upward to the handshake is a configuration change, and the week of airtime is free. An operator who does not — one aircraft flown a few times a month — will never recover the cost of building it, and certificates remain the right answer at any handshake rate. The threshold is a prior decision rather than a fleet size, which is why this is an architectural question.

None of these is an argument against pre-provisioning; they are the reasons it is an architectural decision rather than a configuration flag. They are also the reason the alternative — certificates, at 16.2 s per handshake on a narrowband link once an intermediate is included, against 11.3 s without one — remains the right answer for open fleets, where the operational cost of pairwise key management exceeds the airtime cost of a certificate chain. What this paper adds is that the trade is now quantified: an operator weighing the two is choosing between 3.3 and 11.3 seconds of link time per handshake, or 16.2 with one intermediate, and between a key-distribution problem and a bandwidth problem.

5.5. Relationship to Web-PKI Solutions

The web has the same byte problem and is solving it structurally, by batching many certificates under one signature so that a handshake carries an inclusion proof rather than a chain [23,25]. It is worth noting why that solution does not transfer. Merkle Tree Certificates assume a client that caches landmarks out of band and a certificate authority that issues in bulk; a UAV fleet has neither, and the aggregation is over certificates from one CA rather than over the two endpoints of one session. The UAV analogue of the structural fix is not a Merkle tree but the pre-shared key: both remove the per-session signature, but the pre-shared key removes the certificate as well, which is where 87.9 % of the bytes are.

5.6. Threats to Validity

The link is emulated, not real. The proxy reproduces serialisation and propagation delay faithfully — validated to +0.3 % on a known transfer — but it does not reproduce radio fading, handover, or the queueing behaviour of a real satellite modem. The clean-link numbers are not lower bounds, and the correction is measured rather than assumed: at a 20 % per-segment loss rate (Section 4.7) completion time rises by 1.34 s for the certificate case against 0.45 s for the classical one. The penalty therefore grows with handshake size in absolute terms but shrinks in relative terms, +11.9 % against +22.8 %, which is the opposite of what one would guess, and it does not reorder the architectures. Burstiness is bounded rather than measured: Section 4.7 nets off the two errors, showing that correlation of burst length L divides the mean penalty by L while leaving the ratio between configurations unchanged, and that it lengthens the worst stall from one RTO to $L \times 1.22$ s. So the means here are an upper bound, the tails a lower one, and the ranking holds either way. The link's two other axes are swept rather than assumed: rate over 2.4–38.4 kbps (Section 4.4) and one-way delay over 0–600 ms (Section 4.5), so the single-operating-point objection does not apply to the ratio. What the delay sweep cannot settle is the absolute value of its slope, which is a property of how the emulator charges delay (Section 4.5); only the ratio between configurations is read from it. Nor was the chunk size swept, for the reason Section 3.2 gives: a path that paid delay per smaller unit would move the absolute times up and leave the ratios where they are.

The computation is measured on the host, not on a flight controller. The arithmetic of the handshake was timed on the machine that runs the testbed (Section 3.4), which gives a margin of three orders of magnitude rather than a cycle count for a specific part. A reader who needs the latter has the literature [8,9] and a reason to think it does not change the conclusion: the margin is wider than the gap between any published Cortex-M4

measurement and a desktop core. Where the paper does use embedded figures — the FN-DSA signing cost in Section 4.8 — they are attributed to the literature and used to argue against a design, not to support one.

Two different questions are easy to conflate here, and this paper answers only the first. *Is the arithmetic affordable on a microcontroller?* The margin says yes by three orders of magnitude, and the literature measures it directly. *Can a complete TLS stack run on the flight controller at all?* That is a question about RAM, flash, an operating system abstraction layer and a maintained port, and nothing in this paper bears on it — the stack measured here is a production TLS implementation running on a desktop operating system, and whether it ports to a given avionics platform is the separate question this paper does not address. An operator whose avionics cannot host a TLS stack has a different problem from the one this paper quantifies, and the pre-shared-key recommendation of Section 4.6 is easier to satisfy at the frame layer than at the session layer precisely because of that.

One stack, one version. All measurements use OpenSSL 3.5.7. Another stack may frame records differently, and the 316-byte identity overhead in the pre-shared-key configuration and the 2,200-byte key exchange increment could move by tens of bytes. The 16,726-byte identity increment is determined by the standards and would not move materially.

Single-run sweep. Not every cell was repeated. The certificate sweep of Table 8 carries five runs per row, with standard deviations of 0.024, 0.018, 0.012 and 0.013 s — coefficients of variation of 0.62, 0.20, 0.10 and 0.09 %, the classical row being the least stable for the reason Section 4.1 gives. The rate sweep of Table 9 is two runs per cell and the matrix of Table 5 is single-run, so those cells carry no interval of their own. Elsewhere the five-run spreads of Table 14 are 0.10 % for the certificate handshake, 0.41 % for the pre-shared-key one and 0.67 % for the classical baseline, though the pooled fifteen-run spreads of Section 4.9 are larger — up to 4.6 % — because two runs in forty-five stalled. The smallest effect any table here is asked to resolve is the 1.25× between ML-DSA-65 and ML-DSA-44, a 20 % difference, which is 125 times the pooled coefficient of the configurations being compared and 4.3 times the worst in the batch. Single runs are adequate for the comparisons made at that resolution, and not for finer claims.

The chain depth is one in the main tables. All certificates there are self-signed, so those numbers describe a flat hierarchy. The depth-two case is not extrapolated: it was built and measured, and costs 11,248 bytes and 4.94 s more per handshake, a factor of 1.44 (Section 5.3). Depths above two are a linear extension of that measurement rather than an independent one, and a public PKI with cross-signing or multiple intermediates would sit above our depth-two row. The reader should map our figures onto their own hierarchy rather than treating them as worst cases.

The FN-DSA figure is a prediction. It is labelled as such in Table 13, Figure 7 and the text, and it rests on a slope measured over a range of 11.3 kB that brackets the predicted point. If a reader distrusts the extrapolation, the argument survives on the measured ML-DSA-44 point alone: 9.02 s is already close enough to FN-DSA's predicted 7.88 s that the conclusion — compact signatures are not the lever — does not depend on the prediction.

6. Conclusions

Post-quantum migration of UAV command-and-control links is usually framed as an embedded-computing problem: can the flight controller afford ML-KEM and ML-DSA. Measured on a real TLS 1.3 stack over a 9.6 kbps beyond-line-of-sight link, it is an identity problem. Of the 20,840 bytes a mutual-authentication ML-KEM-768 + ML-DSA-65 handshake puts on the wire, 18,310 are certificate and signature material and 2,530 are key

exchange. The smaller term is 12.1 % of the total and the larger 87.9 %; driving the key exchange to zero — which no algorithm can — would leave the identity term untouched.

The available remedies differ in value by a factor of three and a half. (Times in this section are the single-run values of Tables 8 and 11, which differ from the five-run means of Table 14 by at most 0.02 s; no difference quoted here is affected.) Shrinking the signature scheme from ML-DSA-65 to ML-DSA-44 saves 2.25 s; the extrapolated FN-DSA-512 point saves a further 1.15 s and carries a draft standard, a 2.3× slower signature and 40–57 kB of key state. Removing the certificate entirely, by replacing certificate authentication with a TLS 1.3 external pre-shared key, saves 7.97 s and brings the post-quantum handshake to within 1.67× of its classical cost.

The chain-depth measurement sharpens the same point from the opposite direction. Adding one intermediate certificate costs 11,248 bytes and 4.94 s, which is 2.2 times what the ML-DSA-65 to ML-DSA-44 change saves. An operator who shrinks the signature scheme and then adds an intermediate has made the handshake 2.7 s worse than it started. The lever is which certificates go on the wire, and how many, never how large they are.

Resumption does not escape this. Measured on the same link, a resumed handshake costs 3,335 bytes and 3.51 s against 12,252 and 11.26 for the cold server-authenticated handshake — a third, and a real saving — but it is 433 bytes and 0.20 s *more* than the external pre-shared key, which carries no ticket at all. And in the mutual-authentication configuration that a cross-organisational operator must run, OpenSSL 3.5.7 — a property of this stack, not of the protocol — issues no ticket, so the option is not on the table: a stack that has to authenticate both directions by certificate cannot get there by resuming.

The rate sweep adds a correction that an earlier reading of these numbers would have missed. The 5.66× above is the value at 9.6 kbps; holding propagation delay fixed and moving the rate from 2.4 to 38.4 kbps, the certificate penalty runs from 8.64× to 3.07×. Migration therefore costs most on exactly the links that have the least bandwidth to spare, and the pre-shared-key remedy is worth 4.0× in time at the slow end against 2.5× at the fast end.

The delay sweep closes the symmetric objection. Holding the rate fixed and moving the one-way delay from 0 to 600 ms, completion time is linear in delay for all three configurations, but the certificate handshake pays twice the slope of the classical one — 4.00 against 2.03 s per second of delay — so the absolute gap widens from 8.80 to 9.99 s. The ratio moves the other way, from 7.00× to 4.72×, because latency is an additive term that dilutes a smaller denominator faster than a large one. The net effect is that the geostationary 5.66× is the mildest value any single-hop link will see: a low-Earth-orbit path should expect 6.71× and the zero-latency control 7.00×, and only a two-hop path buys the ratio down. The headline is therefore conservative for any single-hop link in the measured range — 0 to 600 ms of one-way delay, 2.4 to 38.4 kbps — which is the direction an error in a migration study should point.

The design implication is uncomfortable but clear. The intervention that matters is not an algorithm choice that a standards body will make for the operator; it is an architectural choice the operator must make themselves, about whether their aircraft needs to establish sessions with parties they were not provisioned against. Operators who can answer no should pre-provision if they already run a key-distribution mechanism, and be done. Operators who must answer yes should budget eleven seconds per post-quantum handshake on a narrowband satellite TLS link over TCP, and at least that over DTLS — the 4 % figure of Section 3.4 is a framing-only bound and excludes IP and UDP headers and flight-level retransmission, both of which raise it — design their session lifetimes accordingly, and watch the Merkle-tree line of work with interest, because it attacks the same byte budget from a different direction.

Two questions remain open. First, the loss model is independent and memoryless. Section 4.7 bounds what correlation would change — the mean penalty falls as $1/L$ for a burst length L, the ratio between configurations does not move, and a fade of L segments stalls for $L \times 1.22$ s against the 1.2 s this model can produce — but a measured Gilbert–Elliott trace on a real radio is still the honest way to replace that bound. Second, the pre-shared-key recommendation assumes a key distribution mechanism we did not analyse; for a fleet, that mechanism — provisioning, rotation and revocation across aircraft, ground stations and mission participants — may itself be the hard part, and it is a problem in operations rather than in cryptography.

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Abbreviations

The following abbreviations are used in this manuscript:

C2	Command and Control
KEM	Key Encapsulation Mechanism
PQC	Post-Quantum Cryptography
PSK	Pre-Shared Key
UAV	Unmanned Aerial Vehicle

References

1. Shor, P.W. Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer. *SIAM Journal on Computing* **1997**, *26*, 1484–1509. https://doi.org/10.1137/S0097539795293172.
2. National Institute of Standards and Technology. FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard. Technical report, U.S. Department of Commerce, 2024. https://doi.org/10.6028/NIST.FIPS.203.
3. National Institute of Standards and Technology. FIPS 204: Module-Lattice-Based Digital Signature Standard. Technical report, U.S. Department of Commerce, 2024. https://doi.org/10.6028/NIST.FIPS.204.

4. National Institute of Standards and Technology. FIPS 205: Stateless Hash-Based Digital Signature Standard. Technical report, U.S. Department of Commerce, 2024. https://doi.org/10.6028/NIST.FIPS.205.
5. Massimo, J.; Kampanakis, P.; Salter, S. Internet X.509 Public Key Infrastructure: Algorithm Identifiers for ML-DSA. RFC 9881, 2025. https://doi.org/10.17487/RFC9881.
6. Massimo, J.; Kampanakis, P.; Salter, S. Internet X.509 Public Key Infrastructure: Algorithm Identifiers for SLH-DSA. RFC 9909, 2025. https://doi.org/10.17487/RFC9909.
7. The OpenSSL Project. OpenSSL 3.5: ML-KEM, ML-DSA and SLH-DSA Support. https://openssl-library.org/, 2025. Version 3.5.7 used for all measurements.
8. Kannwischer, M.J.; Rijneveld, J.; Schwabe, P.; Stoffelen, K. pqm4: Testing and Benchmarking NIST PQC on ARM Cortex-M4. In Proceedings of the Proceedings of the 2nd NIST PQC Standardization Conference, 2019.
9. Botros, L.; Kannwischer, M.J.; Schwabe, P. Memory-Efficient High-Speed Implementation of Kyber on ARM Cortex-M4. In Proceedings of the Progress in Cryptology – AFRICACRYPT 2019, 2019, Vol. 11627, *Lecture Notes in Computer Science*, pp. 111–130. https://doi.org/10.1007/978-3-030-23696-0_7.
10. Chen, L.; Jordan, S.; Liu, Y.K.; Moody, D.; Peralta, R.; Perlner, R.; Smith-Tone, D. Report on Post-Quantum Cryptography. Technical Report NISTIR 8105, National Institute of Standards and Technology, 2016. https://doi.org/10.6028/NIST.IR.8105.
11. Bernstein, D.J.; Lange, T. Post-Quantum Cryptography. *Nature* **2017**, *549*, 188–194. https://doi.org/10.1038/nature23461.
12. Bernstein, D.J.; Buchmann, J.; Dahmen, E., Eds. *Post-Quantum Cryptography*; Springer: Berlin, Germany, 2009. https://doi.org/10.1007/978-3-540-88702-7.
13. RTCA, Inc.. DO-362: Command and Control (C2) Data Link Minimum Operational Performance Standards (MOPS) (Terrestrial). Technical report, RTCA, Inc., Washington, DC, USA, 2016.
14. Zeng, Y.; Zhang, R.; Lim, T.J. Wireless Communications with Unmanned Aerial Vehicles: Opportunities and Challenges. *IEEE Communications Magazine* **2016**, *54*, 36–42. https://doi.org/10.1109/MCOM.2016.7470933.
15. Mozaffari, M.; Saad, W.; Bennis, M.; Nam, Y.H.; Debbah, M. A Tutorial on UAVs for Wireless Networks: Applications, Challenges, and Open Problems. *IEEE Communications Surveys & Tutorials* **2019**, *21*, 2334–2360. https://doi.org/10.1109/COMST.2019.2902862.
16. Gupta, L.; Jain, R.; Vaszkun, G. Survey of Important Issues in UAV Communication Networks. *IEEE Communications Surveys & Tutorials* **2016**, *18*, 1123–1152. https://doi.org/10.1109/COMST.2015.2495297.
17. MAVLink Development Team. MAVLink Developer Guide. https://mavlink.io/en/, 2024. Accessed: 2026-11-01.
18. Rescorla, E.; Tschofenig, H.; Fossati, T. The Datagram Transport Layer Security (DTLS) Protocol Version 1.3, 2022. https://doi.org/10.17487/RFC9147.
19. Rescorla, E.; Modadugu, N. Datagram Transport Layer Security Version 1.2, 2012. https://doi.org/10.17487/RFC6347.
20. Hussien, A.; Abdullah, A.A. A Post-Quantum Lightweight Mutual Authentication and Encryption Scheme for Secure UAV Communications. *Frontiers in Computer Science* **2026**, *8*. https://doi.org/10.3389/fcomp.2026.1947684.
21. Xia, T.; Wang, M.; He, J.; Yang, G.; Fan, L.; Wei, G. A Quantum-Resistant Identity Authentication and Key Agreement Scheme for UAV Networks Based on Kyber Algorithm. *Drones* **2024**, *8*, 359. https://doi.org/10.3390/drones8080359.
22. MORVS. Swarm SDK 0.4.0: Encrypted Messaging for Drone Swarms. https://morvs.ai/swarm/, 2026. Release specifications and measured message sizes, revised 8 October 2026; accessed 2026-11-01.
23. Benjamin, D.; O'Brien, D.; Westerbaan, B.; Valenta, L.; Valsorda, F. Merkle Tree Certificates. Internet-Draft draft-ietf-plants-merkle-tree-certs-07, 2026. Work in progress, revised 7 October 2026.
24. Chu, J.; Dukkipati, N.; Cheng, Y.; Mathis, M. Increasing TCP's Initial Window. RFC 6928, 2013. https://doi.org/10.17487/RFC6928.

25. Internet Security Research Group. A Post-Quantum Future for Let's Encrypt. https://letsencrypt.org/2026/06/03/pq-future, 2026. Technical announcement, not peer reviewed; published 3 June 2026; accessed 2026-11-01.
26. Sikeridis, D.; Kampanakis, P.; Devetsikiotis, M. Post-Quantum Authentication in TLS 1.3: A Performance Study. In Proceedings of the Proceedings of the 27th Annual Network and Distributed System Security Symposium (NDSS), San Diego, CA, USA, 2020.
27. Sikeridis, D.; Kampanakis, P.; Devetsikiotis, M. Assessing the Overhead of Post-Quantum Cryptography in TLS 1.3 and SSH. In Proceedings of the Proceedings of the 16th International Conference on Emerging Networking Experiments and Technologies (CoNEXT), Barcelona, Spain, 2020; pp. 149–156. https://doi.org/10.1145/3386367.3431305.
28. Paquin, C.; Stebila, D.; Tamvada, G. Benchmarking Post-Quantum Cryptography in TLS. In Proceedings of the Post-Quantum Cryptography – PQCrypto 2020, Cham, Switzerland, 2020; Vol. 12100, *Lecture Notes in Computer Science*, pp. 72–91. https://doi.org/10.1007/978-3-030-44223-1_5.
29. Schwabe, P.; Stebila, D.; Wiggers, T. Post-Quantum TLS Without Handshake Signatures. In Proceedings of the Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security (CCS), Virtual Event, USA, 2020; pp. 1461–1480. https://doi.org/10.1145/3372297.3423350.
30. Rescorla, E. The Transport Layer Security (TLS) Protocol Version 1.3, 2018. https://doi.org/10.17487/RFC8446.
31. Open Quantum Safe Project. liboqs: An Open-Source C Library for Quantum-Safe Cryptography. https://openquantumsafe.org/, 2024. Accessed: 2026-11-01.
32. Pornin, T. New Efficient, Constant-Time Implementations of Falcon. In Proceedings of the Cryptology ePrint Archive, Report 2019/893, 2019.
33. Mosca, M. Cybersecurity in an Era with Quantum Computers: Will We Be Ready? *IEEE Security & Privacy* **2018**, *16*, 38–41. https://doi.org/10.1109/MSP.2018.3761723.